



上海交通大学学位论文

面向 3C 工业生产线的双臂机器人视觉-语言-动作模型

姓名	: 杨皓然	520XXXXXXXXX
	吴律	520XXXXXXXXX
	吴毅昕	520XXXXXXXXX
	程天纵	520XXXXXXXXX
	顾益铭	522370910228

导师 : 班雨桐

指导教师 : 黄佩森

学院 : 浦江国际学院

专业名称 : 电子与计算机工程

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DUAL-ARM ROBOTS IN 3C INDUSTRIAL

PRODUCTION LINES

Name:	Yang Haoran	520XXXXXXXX
	Wu Lu	520XXXXXXXX
	Wu Yixin	520XXXXXXXX
	Cheng Tianzong	520XXXXXXXX
	Gu Yiming	522370910228
Mentor:	Prof. Yutong Ban	
Instructor:	Prof. Peisen Huang	
College:	Global College	

Global College

Shanghai Jiao Tong University

Shanghai, P.R. China

August, 2026

上海交通大学

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摘 要

现代 3C 电子装配线需要处理小型元器件、高产品多样性和富接触操作，这些任务仍难以通过传统固定程序机器人实现自动化。本项目针对这一缺口，面向双臂工业操作开发视觉-语言-动作 (Vision-Language-Action, VLA) 系统。系统设计运行于双臂 Franka 平台，并面向典型 PC 关键部件任务，包括内存条插入、CPU 放置以及电源线连接。

当前部署系统选择相应的 ACT 检查点，接收同步视觉观测和机器人状态，并在闭环执行过程中输出机器人及夹爪动作。现阶段证据限于干运行和有边界的验证活动；本文不声称已经实现语言条件推理、统一多任务控制、自主规划、故障检测、恢复或双臂协同操作。这些功能仍是面向预期 VLA 与双臂系统的长期扩展方向。

工程验收目标包括：规划成功率高于 90%，任务执行成功率高于 70%，部件检测精度高于 90%，位姿误差低于 20 mm，故障检测率高于 90%，恢复成功率高于 70%。这些数值定义验证阈值，而非已经证明的性能结果。

关键词：双臂机器人，视觉-语言-动作模型，工业自动化

ABSTRACT

Modern 3C electronics assembly lines deal with small components, high product variety, and contact-rich operations that remain difficult to automate with conventional fixed-program robots. Our project targets this gap by developing a Vision-Language-Action (VLA) system for bimanual industrial manipulation. The system is designed to run on a dual-arm Franka platform and to handle representative PC key-component tasks—specifically RAM insertion, CPU placement, and power-cable connection.

The deployed system selects the corresponding ACT checkpoint, receives synchronized visual observations and robot state, and outputs robot and gripper actions during closed-loop execution. Current evidence is limited to dry runs and bounded validation activities; the report does not claim demonstrated language-conditioned inference, unified multi-task control, autonomous planning, failure detection, recovery, or bimanual manipulation. These functions remain long-term extensions toward the intended Vision-Language-Action (VLA) and dual-arm system.

The engineering acceptance targets are planning success above 90%, task execution success above 70%, component detection accuracy above 90%, pose error below 20 mm, failure-detection rate above 90%, and recovery success above 70%. These values define validation thresholds rather than demonstrated performance results.

Key words: dual-arm robot, vision-language-action model, industrial automation

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Chapter 1 Introduction

1.1 Background and significance

The 3C industry—computer, communication, and consumer electronics—increasingly demands flexible robotic systems that can cope with short product life cycles, diverse part geometries, small components, and frequent line changeovers. Traditional hard-coded automation approaches are effective for stable, high-volume production, but they exhibit significant limitations in high-mix assembly scenarios. Each product variant typically requires reprogramming, fixture redesign, and system reconfiguration, which increases development and maintenance costs. In contrast, human operators can rapidly understand new tasks from natural-language instructions and adapt to new operating conditions with limited prior programming. This contrast highlights the importance of advancing industrial robotic systems toward more human-aligned capabilities, especially semantic understanding of instructions and data-efficient adaptation under limited demonstrations.

Recent advances in Vision-Language-Action (VLA) models offer a promising route toward this goal. By jointly learning from visual observations, language instructions, robot proprioceptive states, and action data, VLA models can in principle generalize across tasks without requiring separate hand-engineered programs for every product variant. Landmark systems, including RT-2, OpenVLA, RDT-1B, and the π series, demonstrate the potential of transferring semantic knowledge from large vision-language models to physical robot control^[1-4]. Earlier work on visual and language-conditioned manipulation also established complementary design patterns: Transporter Networks use spatially grounded pick-and-place representations, CLIPort combines semantic and spatial pathways, and PerAct couples language goals to structured visual observations^[5-7]. In parallel, BridgeData V2, Open X-Embodiment, DROID, and LIBERO have made dataset scale, embodiment diversity, and reproducible task evaluation central considerations in robot-learning system design^[8-11].

Despite this progress, adapting VLA-style systems to industrial assembly remains a system-level engineering problem. Many published demonstrations emphasize tabletop manipulation, single-arm control, or short-horizon tasks with relatively forgiving tolerances. In contrast, 3C assembly requires reliable perception of small components, accurate contact-rich

manipulation, synchronized sensing, safe robot execution, and repeatable validation under bounded physical conditions. These requirements motivate a complete manipulation pipeline that links hardware selection, teleoperation, data collection, model training, deployment, and quantitative evaluation.

1.2 Design problem

This project develops a dual-arm industrial manipulation platform that couples camera observations, natural-language task instructions, and robot proprioceptive state to learned bilateral actions. The workcell consists of two Franka Research 3 (FR3) robot arms, one fixed overhead camera, two wrist cameras, two custom grippers, and two GELLO leader devices for demonstration collection. Demonstrations are recorded as packet-associated visual observations, bilateral robot and gripper states, absolute target actions, and task instructions.

The implemented task set contains three bimanual manipulation sequences. In the first task, the left arm pushes a can to the middle while the right arm grasps a handled black ring, places it over the can, and releases it. In the second task, the right arm pushes a box containing a power socket to the middle while the left arm grasps a plug and inserts it into the socket. In the third task, the left arm pulls and stabilizes a computer motherboard while the right arm removes the memory module and places it in a yellow tray. The first task serves as a structured reposition–grasp–place benchmark, while the latter two represent constrained connector and computer-component operations.

The final controller is the LeRobot implementation of $\pi_{0.5}$, initialized from the official `pi05_base` checkpoint^[12-15]. One jointly trained checkpoint is shared by all three tasks. At deployment, the same policy receives one of the documented natural-language instructions, three RGB observations, and the current 16-dimensional bilateral state, then predicts absolute targets for both arms and both grippers. The completed implementation therefore provides language-conditioned task input, unified multi-task inference, and dual-arm action generation within the three bounded workcell tasks. It does not claim unrestricted open-world generalization, autonomous task planning, certified failure recovery, or production-line autonomy beyond the evaluated conditions.

1.3 Project objectives and scope

The project objective is to establish an end-to-end engineering pipeline for bounded bimanual industrial-manipulation experiments. This pipeline includes hardware integration, packet-associated multi-camera sensing, bilateral teleoperated demonstration collection, LeRobot Dataset v3.0 preparation, joint multi-task $\pi_{0.5}$ fine-tuning, language-conditioned deployment, and repeatable physical validation of the three scoped tasks.

The engineering acceptance targets inherited from the design-specification stage are planning success above 90%, task execution success above 70%, component detection accuracy above 90%, pose error below 20 mm, failure-detection rate above 90%, and recovery success above 70%. These values define validation thresholds rather than performance claims. Because the implemented $\pi_{0.5}$ policy is an end-to-end controller without separate planning, detection, pose-estimation, or learned recovery modules, Chapter 5 distinguishes directly measurable end-to-end task outcomes from design-stage targets that are not independently observable in the final architecture.

1.4 Main contributions

The main contributions of this thesis are as follows:

1. A dual-FR3 manipulation platform is specified and integrated with GELLO teleoperation, synchronized camera sensing, robot-state recording, and task-compatible end effectors and fixtures.
2. A reproducible data and learning pipeline converts 240 bilateral teleoperated episodes into three LeRobot Dataset v3.0 task datasets with episode-level training and validation splits, equal-probability task sampling, and a common absolute-action representation.
3. A single language-conditioned LeRobot $\pi_{0.5}$ checkpoint is fine-tuned and deployed for three bimanual tasks using three RGB views, a 16-dimensional bilateral state, and a 16-dimensional bilateral action interface.
4. A scoped physical-validation framework defines task-specific terminal conditions, run-termination rules, and end-to-end evaluation procedures for the ring, socket, and memory-module tasks.
5. A traceable engineering process links customer requirements and quality function de-

ployment to concept generation, module selection, implementation decisions, and validation criteria.

1.5 Organization of this thesis

The remainder of this thesis is organized as follows. Chapter 2 translates customer requirements into measurable engineering specifications and presents the quality function deployment. Chapter 3 describes functional decomposition, concept generation, and selection of the major hardware, sensing, teleoperation, end-effector, and policy modules. Chapter 4 details the final system design, data and control architecture, model-training procedure, and implementation plan. Chapter 5 presents the physical validation methods and experimental results. Chapter 6 discusses the strengths and limitations of the system, summarizes the conclusions, and identifies directions for future work.

Chapter 2 Design specification

2.1 Customer requirements

Through discussions with the project sponsor and analysis of the target application, eight customer requirements were identified and assigned importance weights reflecting the sponsor's priorities. The requirements capture both functional goals and non-functional concerns, including robustness, safety, generalizability, cost, and reliability.

Table 2–1 Customer requirements and importance weights

No.	Customer Requirement	Weight
CR1	Complete assembly task successfully	0.18
CR2	High assembly precision	0.16
CR3	Robust to environment and disturbances	0.15
CR4	Safe and collision-free operation	0.14
CR5	Fast task execution	0.12
CR6	Easy to use and generalize	0.10
CR7	Cost-effective system	0.08
CR8	System reliable and recovery capable	0.07

The customer requirements were translated into engineering specifications through the QFD process rather than by assigning targets independently. For each customer requirement i and engineering specification j , the team assigned a relationship value $r_{ij} \in \{0, 1, 3, 9\}$ (none, weak, moderate, or strong) and calculated the unnormalized engineering priority as

$$I_j = \sum_i w_i r_{ij}, \quad (2-1)$$

where w_i is the customer-importance weight in Table 2–1. The priorities were then normalized to identify the technical characteristics that most directly protect the high-weight needs. CR1 and CR5 primarily drive ES1 and ES2; CR2 drives ES3 and ES4; CR3, CR4, and CR8 drive ES5 and ES6; and CR6–CR7 constrain the selected implementation to a usable, feasible, and maintainable route. This translation produces an auditable chain from the sponsor's language to acceptance tests and explains why the later concept search is organized around platform, teleoperation, data collection, gripper, and policy modules.

2.2 Engineering specifications

The customer requirements were converted into six measurable engineering specifications. The targets in Table 2–2 are the final acceptance thresholds for the present proof-of-concept, rather than performance claims. Each rate is calculated over independently initialized physical trials of the relevant task. A trial is counted once only; an unsuccessful recovery is not reclassified as an independent execution attempt. This convention prevents recovery behavior from artificially inflating the reported task-execution success rate.

Table 2–2 Final engineering specifications and acceptance thresholds

No.	Engineering Specification	Target	Operational Definition
ES1	Planning success rate	> 90%	Fraction of trials in which the selected task policy and initialized scene form a valid executable plan.
ES2	Task execution success rate	> 70%	Fraction of trials in which the prescribed assembly operation reaches its task-specific completion condition.
ES3	Component detection accuracy	> 90%	Fraction of evaluated observations in which the required component is correctly detected for the active task.
ES4	Pose estimation error	< 20 mm	Euclidean position error between the estimated and reference component pose.
ES5	Failure detection rate	> 90%	Fraction of induced or naturally occurring execution failures that are identified before further unsafe action is issued.
ES6	Recovery rate	> 70%	Fraction of detected failures for which the defined recovery procedure restores a valid task state.

2.3 Quality function deployment

Figure 2–1 records the resulting House of Quality. Its 9–3–1 relationship scale makes the engineering priorities interpretable rather than decorative: task completion, component localization, pose estimation, and failure handling carry influence across multiple high-weight customer requirements. The roof is used as a design-control device. For example, additional visual coverage can strengthen ES3 and ES4, but it increases calibration, synchronization, and data-management burden; similarly, recovery logic can improve ES5–ES6 while adding

controller integration work. These QFD trade-offs define the subsequent concept space. The platform and gripper must enable precise contact motion for ES1–ES2 and ES4; the tele-operator and data pipeline must produce usable demonstrations and observable scenes; and the policy must convert those observations into safe executable actions while leaving explicit hooks for failure monitoring and recovery.

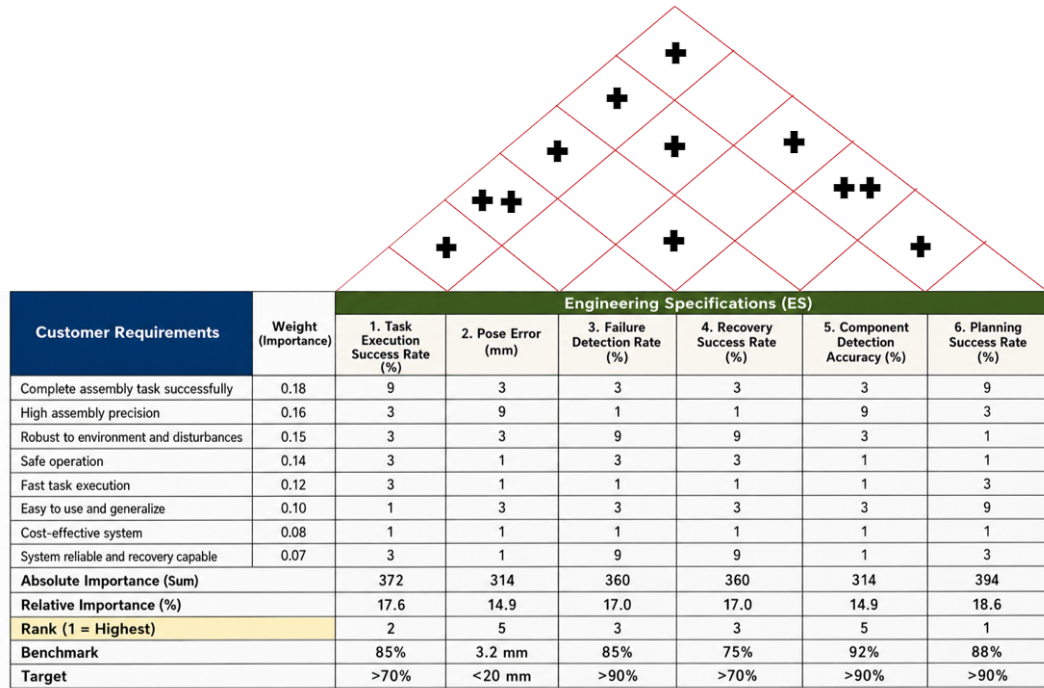


Illustration 2–1 QFD House of Quality for the VLA bimanual manipulation system.

Chapter 3 Concept generation and selection

3.1 Concept generation

3.1.1 Functional decomposition

The target assembly system was first decomposed into solution-neutral functions: establish the workspace state; create a stable robot–operator interface for demonstrations; record synchronized visual, proprioceptive, and action data; grasp and constrain the component during contact; map the current observation to coordinated motion; and detect or recover from an invalid execution state. The decomposition converts the QFD priorities into five design modules. The robot platform provides repeatable bimanual motion for ES1, ES2, and ES4; the teleoperation module enables usable demonstrations; the data-collection module supports ES3–ES4 and learning reproducibility; the gripper realizes component contact; and the policy maps observations and state to actions. Failure monitoring and recovery remain cross-cutting functions because they act across the collection and deployment loop.

3.1.2 Structured brainstorming

The team used structured brainstorming to generate alternatives for each module before choosing a configuration. Candidate ideas were retained when they addressed at least one QFD-linked specification and did not violate laboratory safety, hardware access, or the project schedule. The session considered the available FR3 ecosystem alongside Flexiv and ALOHA-style platforms; direct joint-master, XR, and hand-held teleoperation; progressively richer observation schemas; compliant and rigid end-effectors; and policy families ranging from Diffusion Policy to ACT and future VLA models. This procedure deliberately separates the choice of a data-collection interface from the choice of a learning algorithm, since neither can be justified independently of the sensing and robot-control path.

The generated alternatives were also informed by established research directions rather than by ad hoc feature lists. Robomimic and MimicGen motivate treating demonstration quality and dataset construction as system-level design variables^[16-17]; RVT, 3D Diffusion Policy, and R3M respectively motivate the retained alternatives for multi-view geometry, 3D visuomotor control, and reusable robot visual representations^[18-20]. These references

define the external concept space, while the selected configuration remains constrained by the laboratory hardware, the QFD priorities, and the project schedule.

3.1.3 Morphological chart and generated module concepts

Table 3–1 records the resulting concept space. A row represents one functional module, and a column entry represents one candidate realization. The bold entry in each row is the selected module concept; the table is not a claim that the unselected entries were all built or experimentally benchmarked.

Table 3–1 Morphological chart for the five functional modules

Concept	Candidate A	Candidate B	Candidate C
C1: Platform	Dual Franka FR3	Dual Flexiv platform	ALOHA-style bimanual platform
C2: Teleoperator	GELLO joint-master	PICO 4 Ultra XR	UMI hand-held
C3: Data collection	Two wrist + one top camera + robot state/action	One wrist + one top camera + robot state/action	Robot state/action only
C4: Gripper	Franka parallel gripper	3D-printed hard gripper	3D-printed soft gripper
C5: Policy	ACT in LeRobot	π -style VLA policy	Diffusion policy

The morphological chart produces five leading module concepts, denoted C1–C5. C1 is a dual-FR3 platform; C2 is GELLO-based teleoperation; C3 is a synchronized three-camera, robot-state, and action pipeline; C4 is a 3D-printed soft gripper; and C5 is an ACT policy integrated through LeRobot. Together they form one integrated system concept, rather than five competing end-to-end robot systems. Their alternatives are retained because each exposes a real engineering trade-off: platform precision against integration cost, operator intuitiveness against mapping complexity, observation coverage against synchronization burden, compliance against geometric repeatability, and policy capability against data and deployment maturity. Detailed descriptions of the unselected alternatives are provided in the appendix.

3.2 Concept selection

3.2.1 Selection method and QFD-derived criteria

Selection was conducted in two stages. First, an absolute screen removed a candidate if it could not be made safe in the laboratory, could not interface with the selected robot and data path, or could not be obtained and integrated within the project schedule. Second, the surviving alternatives were scored on a five-point ordinal scale, where 1 denotes a weak fit and 5 denotes a strong fit. Ratings were assigned from the QFD relationship to ES1–ES6, laboratory availability and interface knowledge, the existing collection workflow, and the cited primary or official sources; no rating is presented as an unreported benchmark result. Consistent with the functional decomposition, the “top five concepts” are the five selected module concepts C1–C5. Their combination forms the single system concept carried into the final design. For candidate a in module m , the decision score is

$$S_{m,a} = \sum_{k=1}^4 w_k r_{m,a,k}, \quad (3-1)$$

where $r_{m,a,k}$ is the rating for criterion k and w_k is its QFD-derived weight. Scores are used only to compare alternatives within the same module; they are recorded design judgments, not measured task-performance results.

Table 3–2 QFD-derived criteria for module selection

No.	Criterion	Weight	QFD Link
SC1	Contribution to task performance and precision	0.30	ES1, ES2, ES3, ES4
SC2	Data, control, and software integration compatibility	0.25	ES1, ES2, ES3
SC3	Safety, robustness, and recovery support	0.20	ES5, ES6; CR3–CR4
SC4	Availability, manufacturability, and schedule feasibility	0.25	CR6–CR7 and project plan

3.2.2 Scoring of the five selected module concepts

Table 3–3 provides the detailed selection record for the five leading module concepts. The same criterion definitions and weights are applied in every module so that the scoring

logic remains traceable to the QFD; however, the totals are interpreted only within a module because a robot platform, a gripper, and a policy make different contributions to the integrated system.

Table 3–3 Module-level concept selection matrix (1 = weak fit; 5 = strong fit)

Module	Candidate	SC1	SC2	SC3	SC4	Weighted score
C1 Platform	Dual Franka FR3	5	5	4	3	4.30
	Dual Flexiv	4	3	4	2	3.30
	ALOHA-style bimanual	3	4	3	5	3.85
C2 Teleoperator	GELLO joint-master	4	5	4	5	4.50
	PICO 4 Ultra XR	3	3	3	4	3.25
	UMI hand-held	3	3	3	4	3.25
C3 Data collection	Two wrist + top camera + state/action	5	5	4	3	4.35
	One wrist + top camera + state/action	4	4	4	5	4.20
	Robot state/action only	2	3	3	5	3.25
C4 Gripper	3D-printed soft gripper	4	4	4	5	4.20
	Franka parallel gripper	3	5	4	3	3.70
	3D-printed hard gripper	4	5	3	4	4.05
C5 Policy	ACT in LeRobot	4	5	4	5	4.50
	Diffusion Policy	4	3	4	4	3.75
	π -style VLA policy (future route)	5	2	3	1	2.75

3.2.3 Comparative evaluation and selected concept

C1: Dual Franka FR3 platform. The dual Franka FR3 platform was selected as the mechanical foundation of the final system because it provides repeatable 7-DoF control, a direct connection to the available laboratory control ecosystem, and a credible path toward coordinated bimanual manipulation. Its principal disadvantage is the substantial integration effort required for workspace coordination, collision avoidance, commissioning, and bilateral control. Nevertheless, these costs are justified by the project objective: lower-cost alternatives

may reduce initial hardware burden, but they would introduce a different control ecosystem or provide a weaker fit to the required assembly precision.

C2: GELLO teleoperation. GELLO was selected as the planned demonstration-collection interface because it offers a direct and intuitive connection between operator motion and the robot-state/action records required by the learning pipeline^[21]. Despite the need for reliable leader-follower mapping and calibration, GELLO remains preferable to immersive or hand-held alternatives for the present system because it provides the most direct route from tele-operated demonstrations to the joint-space data representation used by the FR3 and LeRobot workflow.

C3: Three-camera data pipeline. The selected sensing configuration uses one Intel RealSense D435 overhead camera and two Intel RealSense D405 wrist-mounted cameras, together with synchronized robot state, gripper state, and action records. The D435 provides a stable global view of the shared workspace, while the two D405 cameras preserve local visual information near the end-effectors and contact regions. This complementary arrangement reduces visual blind spots and directly supports the component-detection and pose-estimation requirements in ES3–ES4, at the cost of increased synchronization and data-management effort.

C4: 3D-printed soft gripper. The selected end-effector is a 3D-printed soft gripper derived from the open-source gripper design in the referenced UMI material; the project adopts only the gripper-finger component rather than reproducing the complete UMI hardware system^[22]. Compliant contact and rapid geometry iteration are valuable for handling assembly components with small geometric variation, although deformation may reduce pose repeatability compared with a rigid locating gripper.

C5: ACT policy in LeRobot. ACT was selected as the current policy because it is integrated with the LeRobot workflow and supports task-specific training and checkpoint deployment from synchronized visual observations and robot-state inputs^[14,23]. Action chunking provides a practical temporal-control mechanism for the present imitation-learning pipeline. Deployment remains task-specific and operator-selected: the policy does not yet infer tasks from natural-language input or demonstrate unified multi-task control. This limitation is accepted because ACT provides a deployable and traceable baseline for the current project stage, whereas Diffusion Policy remains an earlier baseline and π -style VLA models remain

a future extension.

The selected architecture is the combination of the highest-scoring surviving candidate in each module. C1 and C4 establish the physical capability for controlled contact; C2 and C3 make demonstrations and visual evidence available to the learner; and C5 provides the deployable policy path. This combined decision covers the full ES1–ES6 chain while retaining a feasible implementation route for subsequent final design, manufacturing, and validation.

Chapter 4 Final design and implementation

4.1 Selected concept description

4.1.1 Design evolution from the DR3 baseline

The final implementation retains the modular physical architecture selected in Design Review 3 (DR3), but the learning system and demonstrated scope have changed substantially. DR3 proposed a dual-Franka Research 3 (FR3) workcell, GELLO demonstration collection, two wrist cameras and one overhead camera, custom compliant fingers, a LeRobot dataset, and an imitation-learning policy. At that stage, Action Chunking with Transformers (ACT) and task-specific checkpoints were the deployable baseline. Language-conditioned bimanual control was a future objective.

The completed system replaces the task-specific ACT controller with the LeRobot implementation of $\pi_{0.5}$. It uses two FR3 arms, two grippers, three RGB views, a 16-dimensional bilateral state and action interface, and one jointly trained multi-task checkpoint. The three tasks are selected by natural-language instructions rather than by manually loading three independent policy files. ACT therefore remains part of the design history and concept-selection evidence, but it is not presented as the final controller.

The mechanical elements selected in DR3 remain directly relevant. The two FR3 arms provide a common kinematic and controller interface, and the two GELLO leaders provide matched bilateral demonstration inputs. The wrist and overhead cameras provide complementary local and global views. The 3D-printed compliant fingers provide a task-compatible contact interface. The later policy update changes how these modules are connected and trained, but does not remove the need for their original engineering analysis.

4.1.2 Overall system architecture

The final system is organized as four connected layers:

1. the physical workcell, consisting of two FR3 arms, two custom grippers, task fixtures, and three cameras;
2. the demonstration layer, consisting of the two GELLO leader devices, the ROS 2 bridge, and the episode recorder;

3. the learning layer, consisting of LeRobot Dataset v3.0, the $\pi_{0.5}$ training configuration, and task-balanced validation; and
4. the deployment layer, consisting of language-conditioned policy inference, action-chunk handling, the bilateral robot adapter, and deterministic safety checks.

During demonstration collection, the operator moves the two GELLO leaders. The follower-side system commands the two FR3 arms. It records three RGB images, the measured bilateral state, the bilateral arm targets, the gripper commands, the episode index, and the task instruction. Accepted episodes are stored in LeRobot Dataset v3.0 format. During training, the three task datasets are sampled with equal task probability and used to update one $\pi_{0.5}$ model. During deployment, one of the three language instructions and the current three-view observation are supplied to the same checkpoint. The policy returns an action chunk, and the robot-side adapter sends the admissible absolute targets to the two FR3 controllers.

Figure 4–1 is retained from DR3 because it records the selected modular architecture and the intended connection between demonstration collection, learning, and closed-loop execution. Two parts of the diagram are historical: the ACT block has been replaced by $\pi_{0.5}$, and the task-specific checkpoint block has been replaced by one unified checkpoint. The three-camera workcell, GELLO collection concept, LeRobot dataset, action-chunk interface, dual-FR3 controller, and deterministic failure-handling boundary remain consistent with the final system.

4.1.3 Selected module concepts

4.1.3.1 Dual-FR3 bimanual platform

Two FR3 arms are used as the follower robots. The use of the same robot model on both sides reduces interface asymmetry: each side contributes seven joint positions, receives seven absolute joint targets, and uses the same low-level controller family. The shared kinematic structure also makes bilateral indexing, joint-limit validation, and data normalization easier to audit. Bimanual operation is required because each task includes a preparation or stabilization action followed by a more precise manipulation action.

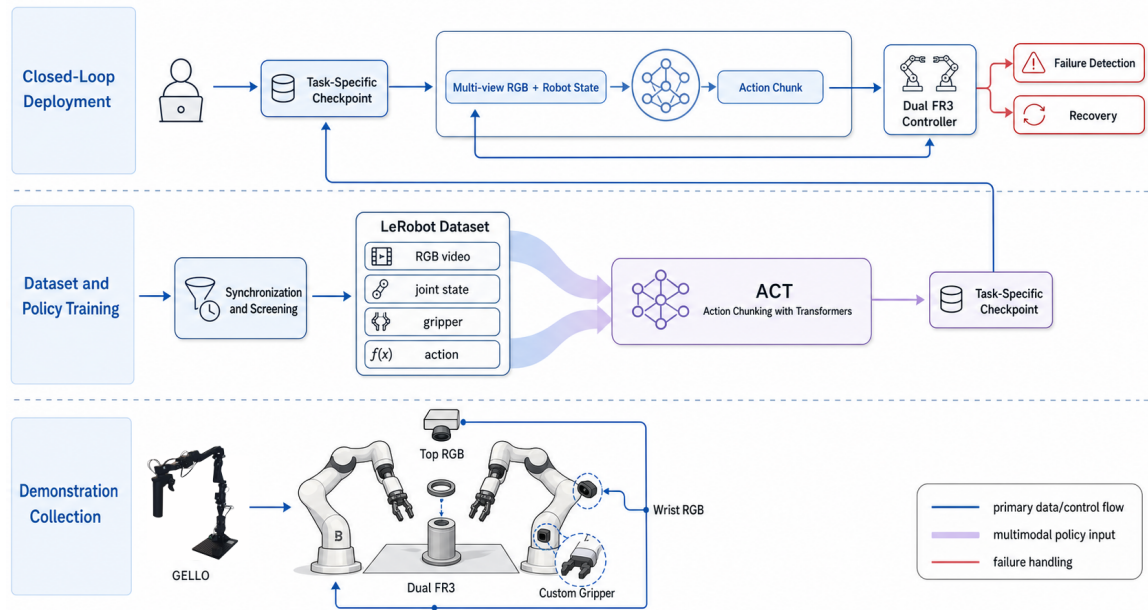


Illustration 4–1 System architecture retained from DR3 as the design baseline. In the final implementation, the ACT and task-specific-checkpoint blocks are superseded by one language-conditioned LeRobot $\pi_{0.5}$ checkpoint; the physical workcell and data-flow modules remain applicable.

4.1.3.2 GELLO demonstration interface

The demonstration interface uses one GELLO leader for each follower arm. GELLO provides a physical leader whose joint arrangement is mapped to the follower robot, giving the operator direct control of coordinated arm motion^[21]. The selected project configuration provides seven arm channels and one gripper channel per leader. Its low-cost modular construction also allows damaged or geometrically unsuitable leader components to be reprinted without changing the follower robot.

4.1.3.3 Three-camera visual observation

The final observation contains one fixed overhead view and two wrist views. The overhead Intel RealSense D435i observes the shared workspace, fixture positions, and large object motion. The two Intel RealSense D405 cameras move with the left and right wrists and preserve local visual information near grasp, contact, insertion, extraction, and placement regions. This arrangement avoids relying on a single global view that may be occluded by

either arm, while avoiding a wrist-only configuration that loses global task context.

4.1.3.4 Custom compliant grippers

Each arm uses the custom gripper developed during DR3. An assembly contains two identical screw-mounted fingers printed in TPU 64D. Each finger has an overall bounding-box size of

$$123.36 \times 25.80 \times 40.00 \text{ mm.}$$

The tapered outline and repeated ribs permit passive deformation around the handled ring, the insulated plug body, the motherboard edge, and the memory module. The reinforced mounting end keeps the screw interface stiffer than the contact region. Compliance is used to increase geometric tolerance; it is not treated as a substitute for camera coverage, accurate demonstration, or robot-side contact limits.

4.1.3.5 LeRobot $\pi_{0.5}$ policy

The final policy is the LeRobot implementation of $\pi_{0.5}$, adapted from the open-source OpenPI implementation^[12-15]. It receives three RGB images, the current bilateral state, and a language instruction. It predicts a chunk of bilateral actions in the same absolute joint-and-gripper representation used by the recorder and executor. The same model parameters are used for all three tasks.

4.1.4 Operational workflow

4.1.4.1 Workcell initialization

Before an episode or an autonomous trial begins, the two arms are returned to the documented initial joint region, the grippers are inspected, and the task objects are reset. Camera mounts, image identities, network connections, controller status, workspace boundaries, and emergency-stop access are checked. The socket fixture is kept electrically unpowered. A run is not started if a camera is missing, a robot state is stale, or the scene does not match the selected instruction.

4.1.4.2 Demonstration collection

The operator performs the selected task using the two GELLO leaders. The ROS 2 bridge assembles packets containing the three images, the measured robot and gripper state, and the

commanded arm and gripper values. The recorder samples these packets at 5 Hz and stores a complete episode. The operator then reviews task completion and resets the workcell before the next episode. Episode boundaries are preserved so that complete trajectories, rather than individual frames, can be assigned to training or validation.

4.1.4.3 Dataset screening and training

The recorded episode is checked for the presence and consistent length of all three image streams, the 16-dimensional state, the 16-dimensional action, the task identity, and the episode metadata. Accepted episodes are added to one of the three task datasets. Training then samples the three tasks with equal probability, applies the defined image and numeric preprocessing, and fine-tunes one $\pi_{0.5}$ checkpoint for 100,000 optimizer steps.

4.1.4.4 Closed-loop deployment

At deployment, the operator selects the natural-language instruction that matches the prepared scene. The model receives the instruction, three current images, and the current bilateral state. It predicts 50 actions, of which the first 10 are consumed before a new observation is evaluated. The executor checks the returned targets and sends them to the two arms at the dataset-aligned control rate. A task terminates on completion, timeout, invalid observation, invalid action, protective stop, emergency stop, or manual intervention.

4.2 Engineering design analysis

4.2.1 Relevant engineering fundamentals

The implemented system combines serial-manipulator kinematics, shared-workspace bi-manual coordination, compliant contact, multi-view machine perception, imitation learning, packet-level data synchronization, and receding-horizon control. The learned policy produces high-level position targets. It does not replace the FR3 low-level servo controller, joint-limit enforcement, protective-stop behavior, or the external emergency-stop procedure.

For one seven-degree-of-freedom arm with joint vector

$$\mathbf{q}_t = [q_{1,t}, \dots, q_{7,t}]^T,$$

the end-effector pose is

$${}^B\mathbf{T}_{E,t} = f(\mathbf{q}_t), \quad (4-1)$$

where B is the robot-base frame, E is the end-effector frame, and $f(\cdot)$ is the forward-kinematics map. In the dual-arm system, the corresponding transforms are

$${}^{B_L}\mathbf{T}_{E_L,t} = f_L(\mathbf{q}_t^L), \quad {}^{B_R}\mathbf{T}_{E_R,t} = f_R(\mathbf{q}_t^R). \quad (4-2)$$

If both base poses are expressed in a common workcell frame W , the relative end-effector transform can be written as

$${}^{E_L}\mathbf{T}_{E_R,t} = ({}^W\mathbf{T}_{B_L} {}^{B_L}\mathbf{T}_{E_L,t})^{-1} {}^W\mathbf{T}_{B_R} {}^{B_R}\mathbf{T}_{E_R,t}. \quad (4-3)$$

Equation 4–3 explains why the two base installations and task fixtures must remain fixed: the learned policy observes images and joint states rather than an explicitly recalibrated relative pose at every episode.

4.2.2 Design assumptions and constraints

The system operates in a structured indoor workcell. Lighting, camera mounts, robot bases, and task fixtures remain sufficiently repeatable for the learned visual policy. Each trial begins within the demonstrated workspace. The can, handled ring, socket box, plug, motherboard, memory module, and yellow tray are treated as task-specific laboratory objects rather than arbitrary unseen industrial components.

The current model uses RGB images and bilateral proprioception. Although the selected RealSense devices can provide depth, and the D435i includes inertial sensing, depth and inertial measurements are not included in the implemented $\pi_{0.5}$ feature contract. The chapter therefore uses manufacturer depth ranges and fields of view to justify camera placement, but it does not claim that the policy performs explicit metric depth reconstruction or six-degree-of-freedom pose estimation.

The recorder relies on software packets assembled by an upstream ROS 2 bridge. It does not use a hardware trigger or exposure-level synchronization. It also does not perform device-time interpolation or ROS approximate-time alignment. The valid engineering claim is packet-level synchronization, not exact simultaneity of the three camera exposures.

The design also assumes that deterministic safeguards remain outside the policy. The learned action is accepted only after dimensional, finite-value, joint, gripper, workspace, freshness, and communication checks. The socket task uses an electrically unpowered fixture.

The system is a supervised research workcell and is not represented as a certified production cell.

4.2.3 FR3 parameters and bimanual kinematic suitability

Each FR3 provides seven degrees of freedom, a nominal payload of 3 kg, a maximum reach of 855 mm, and link-side torque sensing in all seven axes^[24-26]. These values are suitable for the project because the manipulated objects are light, the task is performed on a tabletop, and the seventh joint gives redundancy for changing wrist orientation while avoiding fixtures and the other arm.

Table 4–1 FR3 parameters relevant to the selected bimanual workcell

Parameter	Value	Design relevance
Degrees of freedom	7 per arm	Provides redundant approach configurations and wrist orientation control.
Nominal payload	3 kg	Exceeds the mass required for the project objects and custom fingers.
Maximum reach	855 mm	Supports a shared central tabletop region when the two bases are arranged on opposite sides.
Joint sensing	Link-side torque sensing in all seven axes	Supports compliant low-level control, contact monitoring, and protective behavior.

The complete joint ranges, mounting specification, environmental limits, controller interfaces, and dimension drawings are reference data rather than design arguments. They are collected in Appendix 2, Table 2–1, and Figures 2–1–2–2. The practical task workspace is deliberately smaller than those hardware bounds so that action chunks do not intentionally approach joint limits.

Figure 4–2 is retained in the main analysis because it directly supports placement of the shared task region. The 855 mm maximum reach is not treated as an invitation to place objects at the boundary: motion near maximum extension reduces available orientation freedom and can bring the arm closer to singular configurations.

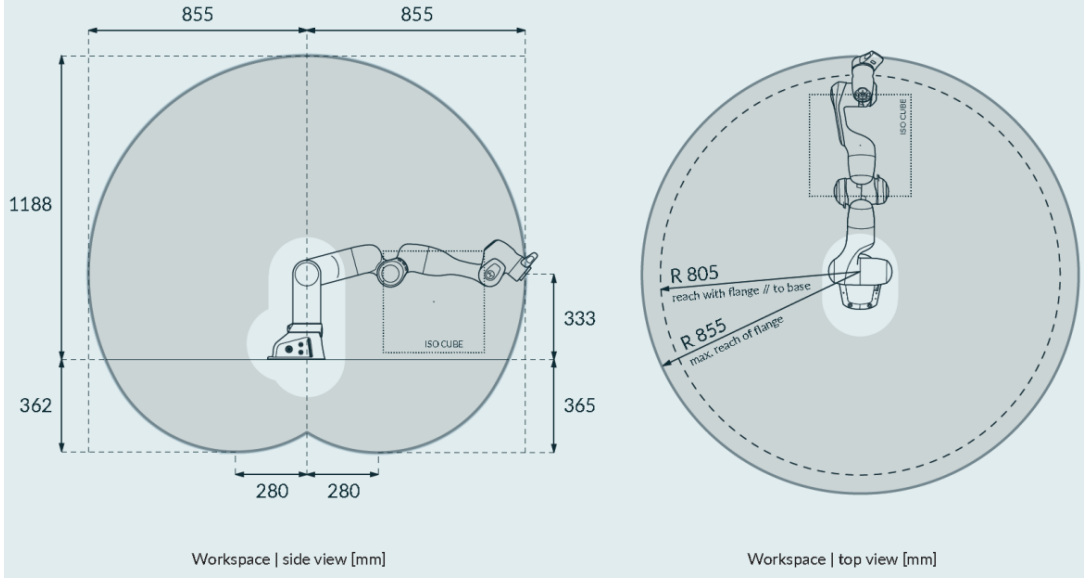


Illustration 4-2 Side and top views of the FR3 workspace envelope. The shared task region is selected inside the overlap of the two practical arm workspaces rather than at the 855 mm reach boundary.

4.2.4 Bimanual state and action representation

The actual state and action are both stored as 16-dimensional `float32` arrays. The state is

$$\mathbf{s}_t = [q_{1,t}^L, \dots, q_{7,t}^L, g_t^L, q_{1,t}^R, \dots, q_{7,t}^R, g_t^R]^T \in \mathbb{R}^{16}, \quad (4-4)$$

where g_t^L and g_t^R are measured gripper opening widths in metres. The action uses the same ordering:

$$\mathbf{a}_t = [q_{1,t+1}^{L,target}, \dots, q_{7,t+1}^{L,target}, g_t^{L,target}, q_{1,t+1}^{R,target}, \dots, q_{7,t+1}^{R,target}, g_t^{R,target}]^T. \quad (4-5)$$

Indices 0–6 correspond to the left arm, index 7 to the left gripper, indices 8–14 to the right arm, and index 15 to the right gripper. Arm actions are absolute next-step joint targets, not joint increments. Gripper actions are absolute `open_width_m` commands. The implemented setting is therefore `use_relative_actions=false` in both training and deployment.

The $\pi_{0.5}$ implementation pads state and action features to an internal maximum of 32 dimensions. Only entries 0–15 contain project data. Entries 16–31 are padding and are never interpreted as additional robot commands. The robot adapter explicitly slices and validates

the first 16 entries before decomposing them into left-arm, left-gripper, right-arm, and right-gripper targets.

4.2.5 Shared workspace and role allocation

The two arms operate in a shared central region, so task role allocation is selected to reduce unnecessary crossing. In Task 1, the left arm pushes the can toward the centre while the right arm grasps and places the handled ring. In Task 2, the right arm pushes the socket box while the left arm grasps and inserts the plug. In Task 3, the left arm grips and pulls the motherboard while the right arm removes the memory module and places it in the yellow tray.

These assignments separate the large preparation motion from the subsequent precision motion. They also preserve the view of the precision-side wrist camera: the wrist that performs the ring placement, plug insertion, or RAM extraction approaches the relevant contact region directly. Role allocation is encoded by the demonstrations and the fixed action indexing; it is not dynamically reassigned by a separate planner.

The workcell dimensions and exact base-to-fixture transforms were not supplied as final measured values. Consequently, this report does not invent a dimensioned collision-free volume. The implemented engineering rule is that fixtures are placed in an experimentally verified overlap region, nominal trajectories stay away from the manufacturer reach boundary, and the robot-side workspace limits are tighter than the theoretical workspace shown in Figure 4–2. A production release would require the final base poses, fixture coordinates, swept-volume analysis, and inter-arm collision geometry to be documented numerically.

4.2.6 Contact, clearance, and insertion tolerance analysis

4.2.6.1 Ring-over-can placement

The ring task contains three mechanically distinct phases: a planar push of the can, a grasp of the ring handle, and a vertical placement of the ring over the can. Successful placement requires the ring inner opening to remain clear of the can outer profile throughout descent. If D_r is the effective ring inner diameter and D_c is the effective can outer diameter, the ideal radial clearance is

$$c_r = \frac{D_r - D_c}{2}. \quad (4-6)$$

The allowable lateral placement error must be smaller than c_r , with an additional margin for ring tilt, finger deformation, and uncertainty in the apparent object boundary. The actual D_r and D_c values were not provided, so Equation 4–6 defines the required measurement rather than claiming a numerical tolerance.

The handle allows the right gripper to avoid pinching the ring circumference, but it introduces an offset between the grasp point and the ring centre. This offset creates a moment that can tilt the ring during transport. A near-vertical final approach and release after the ring is supported by the target reduce the sensitivity to that moment.

4.2.6.2 Plug insertion

Plug insertion is a constrained contact task. Lateral and angular misalignment can produce edge contact, jamming, incomplete seating, socket-box displacement, or excessive force. For an insertion depth L_i and effective lateral clearance c_i , a first-order small-clearance angular bound is

$$\theta_{\max} \approx \tan^{-1} \left(\frac{c_i}{L_i} \right). \quad (4-7)$$

The relation shows that angular tolerance decreases as insertion depth increases. The socket box is pushed into a repeatable central region before insertion so that the left arm can approach with sufficient remaining wrist orientation range. The right-arm push must end before the left-arm precision approach enters the same local region.

The DR3 socket mock-up was electrically unpowered and had a recorded overall size of $80 \times 80 \times 30$ mm. The current task continues to use an unpowered socket fixture. The exact dimensions of the final socket box, pin clearance, insertion depth, and fixture compliance should be remeasured if the physical object differs from the DR3 mock-up. Until those values are recorded, success is defined functionally: the correct plug enters the correct socket, remains seated without manual assistance, and produces no visible damage.

4.2.6.3 Memory-module extraction

The motherboard task requires fixture stabilization, connector release, module extraction, and safe placement. The left gripper pulls the motherboard closer and holds or stabilizes it. The right gripper must approach the memory module without contacting nearby components, apply a controlled extraction motion consistent with the connector geometry, and transport the released module to the yellow tray.

The mechanical tolerance is governed by the free space around the memory module, the width and compliance of the custom fingers, and the connector's extraction direction. Excess lateral motion can bend the module or load adjacent board components. Because connector retention force, neighbouring-component clearance, and module dimensions were not supplied, the report does not infer extraction force or a numerical clearance. These quantities remain required measurements for a complete contact-mechanics verification.

4.2.7 Custom gripper geometry and mechanical analysis

Each gripper assembly uses two identical TPU 64D fingers, and the dual-arm workcell therefore uses four fingers. For one finger,

$$L = 123.36 \text{ mm}, \quad T = 25.80 \text{ mm}, \quad H = 40.00 \text{ mm}.$$

The principal geometric ratios are

$$\frac{L}{T} = \frac{123.36}{25.80} \approx 4.78, \quad \frac{L}{H} = \frac{123.36}{40.00} \approx 3.08. \quad (4-8)$$

These ratios describe an elongated finger rather than a short rigid jaw. The repeated ribs reduce local bending stiffness along the contact region, the tapered outline improves access, and the thicker mounting end distributes screw load.

The same compliance has a trade-off. It accommodates object-shape and pose variation and can reduce concentrated contact pressure, but deformation can reduce grasp repeatability and make the effective fingertip pose load dependent. The task design therefore uses compliance for tolerance while retaining fixed camera identities, repeatable starts, and arm-side position control. No finite-element stress, fatigue-life, or calibrated force-deflection result is claimed because print orientation, infill, wall count, actual TPU properties, contact loads, and cycle count were not supplied. The full orthographic drawing and fabrication-reference dimensions are provided in Appendix 2, Figure 2–4.

4.2.8 Perception working distance, field of view, and camera selection

The overhead camera is an Intel RealSense D435i. Its specified ideal depth range is approximately 0.3–3 m and its depth field of view is approximately $87^\circ \times 58^\circ$ ^[27]. This makes it suitable for observing the complete tabletop from a fixed elevated position. The overhead

view supplies global information when either wrist view is occluded or too close to show the relationship between the object, fixture, and destination.

Each wrist camera is an Intel RealSense D405. Its specified ideal depth range is approximately 7–50 cm, with an approximately $87^\circ \times 58^\circ$ depth field of view^[28]. This range is better matched to a camera mounted near the end effector. The wrist views preserve local detail during ring descent, plug alignment, and memory-module extraction.

Table 4–2 Camera configuration and selection basis

Camera	Quantity	Nominal range	ideal	Depth FOV	Project function
RealSense D435i	1	0.3–3 m		$87^\circ \times 58^\circ$	Fixed overhead context for the shared workcell, fixtures, destinations, and large object displacement.
RealSense D405	2	7–50 cm		$87^\circ \times 58^\circ$	Left- and right-wrist close-range views for grasp, contact, insertion, extraction, and placement.

The hardware can operate at higher image and depth rates than the learning pipeline, but the project records RGB at 424×240 and 5 Hz. The selected policy does not consume depth or D435i inertial data. Camera range and field of view therefore justify coverage and mounting, while the learned input remains three RGB images.

4.2.9 Image format and preprocessing

All three raw streams have width 424 and height 240. The LeRobot dataset stores each decoded RGB image in channel-first form:

$$[C, H, W] = [3, 240, 424].$$

Before an image is passed to $\pi_{0.5}$, it is converted to a $3 \times 224 \times 224$ tensor. The conversion preserves aspect ratio. The 424×240 image is scaled to approximately 224×126 , and

the remaining square area is filled with approximately 49 black pixels above and below the resized image. The image is not stretched to 224×224 .

Pixel values are represented in $[0, 1]$ after decoding and are normalized to $[-1, 1]$ before the model. This process is applied independently but identically to the overhead, left-wrist, and right-wrist images. Camera names and ordering are part of the feature contract. A swapped wrist-camera mapping can produce a dimensionally valid but semantically incorrect policy input, so identity is checked before recording and deployment.

4.2.10 GELLO channel architecture and leader–follower mapping

The bilateral collection system uses two GELLO–Franka leaders. Each leader provides seven arm channels and one gripper-related channel, matching the seven joint targets and one gripper command required by its FR3 follower. This channel correspondence is the design-relevant property; the detailed actuator, controller-board, power-supply, structural-kit, and dual-system quantities are listed in Appendix 2, Table 2–2. The leader drawing and actuator reference data are also moved there.

Leader motion must be mapped to follower joint order, sign convention, zero offset, and permitted range. An incorrect offset can generate a valid numeric command that is unsafe for the follower. The pre-collection procedure therefore verifies the leader zero, moves each channel through a small range, confirms the corresponding follower direction, and checks the gripper-width mapping before a task episode is recorded.

4.2.11 Data-collection pipeline

The implemented data path is

two GELLO leaders → two FR3 follower controllers → ROS 2 bridge
 → assembled packets
 → LeRobot recorder → Parquet records and three RGB videos.

The follower-side system provides measured joint positions and current gripper widths. The command side provides absolute arm targets and absolute gripper-width commands. The recorder samples the assembled packet stream at 5 Hz, assigns frames to the active episode, and preserves the task identity. A complete episode contains a continuous visual and propri-

oceptive record from the initial condition to either task completion or termination.

Collection and task review are intentionally separate operations. The recorder verifies schema and continuity, while the operator verifies whether the physical task was demonstrated correctly. A numerically complete episode can still be unsuitable for training if the operator intervened incorrectly, the object left the intended workspace, or the terminal state does not match the language instruction.

4.2.12 Packet synchronization and action-label alignment

The stored cadence is 5 Hz, corresponding to approximately 0.2 s per step. The recorder pairs adjacent assembled packets. For the saved sample at index t ,

$$\mathbf{o}_t^{\text{visual}} = \text{three images from packet } t, \quad (4-9)$$

$$\mathbf{s}_t = \text{bilateral measured state from packet } t, \quad (4-10)$$

$$\mathbf{a}_t^{\text{arm}} = \mathbf{q}_{t+1}^{\text{target}} \text{ from packet } t + 1, \quad (4-11)$$

$$\mathbf{a}_t^{\text{gripper}} = \mathbf{g}_t^{\text{target}} \text{ from packet } t. \quad (4-12)$$

Images and measured state therefore come from the same current packet. The arm label is shifted forward by one packet so that the current observation is paired with the next absolute joint target. The gripper label follows the existing same-packet convention.

This procedure guarantees consistency of the saved software packet and label rule. It does not prove that all three camera exposures occurred at the same physical instant. The recorder does not interpolate device timestamps, perform ROS approximate-time alignment, or use a hardware trigger. It trusts the upstream bridge to provide an assembled packet. This limitation must be retained in the report because a nominal 5 Hz LeRobot timestamp alone cannot be used to reconstruct true camera skew or end-to-end latency.

4.2.13 Dataset construction and episode split

The three task datasets contain 80 episodes each. All three use random seed 1000 and the same validation episode indices:

$$\{17, 35, 58, 65, 66, 72, 74, 75\}.$$

The remaining 72 episodes per task are used for training. Splitting by episode prevents neighbouring frames from the same physical trajectory from appearing in both the training and

validation sets.

Table 4–3 Episode-level dataset split for the three bimanual tasks

Task dataset	Episodes	Frames	Training split	Validation split
dual_ring	80	46,326	72 episodes / 41,410 frames	8 episodes / 4,916 frames
dual_socket	80	38,511	72 episodes / 34,655 frames	8 episodes / 3,856 frames
dual_ram	80	34,813	72 episodes / 31,675 frames	8 episodes / 3,138 frames
Total	240	119,650	216 episodes / 107,740 frames	24 episodes / 11,910 frames

The tasks are sampled with equal probability during training. Task selection occurs before sample selection, so `dual_ring` does not receive a larger task-level probability merely because it contains more frames. This choice balances the three instructions and prevents the longest average trajectory from dominating the gradient updates.

4.2.14 $\pi_{0.5}$ base model and software configuration

The final controller uses the official `pi05_base` model. The upstream checkpoint is identified by `gs://openpi-assets/checkpoints/pi05_base/params`, and training reads the converted PyTorch parameters from `real-exp/.hf-cache/pi05_base_pytorch/model.safetensors`. The base architecture combines a PaliGemma `gemma_2b` vision-language component with a `gemma_300m` action expert. The model weights use `bfloat16`.

The project uses LeRobot 0.5.2 with local project modifications. The base commit is

`dc06dba9d6220ab99c94e83ec7f0d45a35b44bed`.

The dataset format is LeRobot Dataset v3.0. Training is executed on physical GPU 1 of an NVIDIA RTX A6000 with 48 GB of memory.

4.2.15 Training procedure

The three tasks are trained jointly for 100,000 optimizer steps. Batch size is 4 and the optimizer is AdamW. The peak learning rate is 2.5×10^{-5} . A 1,000-step warm-up is followed by cosine decay to 2.5×10^{-6} . Task-balanced validation is run every 5,000 steps.

Table 4–4 Final $\pi_{0.5}$ training configuration

Parameter	Implemented value
Base model	Official pi05_base
Model structure	PaliGemma gemma_2b and action expert gemma_300m
Weight precision	bfloat16
LeRobot version	0.5.2, with local modifications
Dataset format	LeRobot Dataset v3.0
Training hardware	NVIDIA RTX A6000 48 GB, physical GPU 1
Batch size	4
Optimizer	AdamW
Training length	100,000 joint-training steps
Learning-rate schedule	1,000-step warm-up to 2.5×10^{-5} , followed by cosine decay to 2.5×10^{-6}
Task sampling	Equal probability for dual_ring, dual_socket, and dual_ram
Validation interval	Every 5,000 optimizer steps

The training configuration deliberately separates task balancing from frame count. The action and state normalizers are derived from the training data only. Validation episodes are not used to compute model updates. The same action representation and `use_relative_actions=false` setting are retained when the selected checkpoint is loaded for deployment.

4.2.16 Validation and checkpoint selection

For each task, validation first computes the mean normalized flow-matching loss over at most 64 batches. With batch size 4, this corresponds to at most 256 validation samples per task at one validation point. The checkpoint-selection quantity is the unweighted task macro-average

$$\mathcal{L}_{\text{macro}} = \frac{\mathcal{L}_{\text{ring}} + \mathcal{L}_{\text{socket}} + \mathcal{L}_{\text{ram}}}{3}. \quad (4-13)$$

The best checkpoint is the validation point with the lowest $\mathcal{L}_{\text{macro}}$.

This metric is intentionally not weighted by the number of validation frames. It gives each task equal influence, just as the training sampler gives each task equal selection probability. The loss supports checkpoint comparison, but it is not a physical success rate, insertion-depth

measurement, contact-force measurement, or calibrated pose error. Real-robot performance is evaluated separately in Chapter 5.

4.2.17 Action chunking and temporal interface

Collection and execution use 5 Hz, so one action step represents 0.2 s. The model predicts a 50-step chunk:

$$T_{\text{predicted}} = \frac{50}{5} = 10 \text{ s.} \quad (4-14)$$

Deployment uses `n_action_steps=10`, so the executor consumes approximately

$$T_{\text{executed}} = \frac{10}{5} = 2 \text{ s} \quad (4-15)$$

before requesting a new policy result from updated observations.

This is a receding-horizon interface. The 50-step prediction gives the action expert a longer temporal target, while executing only 10 steps limits the amount of open-loop motion. The exact policy-query interval may also include capture, preprocessing, inference, transfer, and queue overhead; therefore, the nominal two-second value describes the consumed action segment rather than a measured end-to-end latency.

4.2.18 Deployment pipeline

Deployment begins by loading the selected unified checkpoint and its associated normalization statistics. The system verifies the three camera feature names, checks the 16-dimensional state, and confirms the language instruction. The current observation is preprocessed and passed to the policy server. The returned action chunk is denormalized, sliced to the 16 valid robot dimensions, checked for finite values and limits, and divided into left-arm, left-gripper, right-arm, and right-gripper targets.

The robot adapter sends absolute joint targets and absolute gripper widths. It does not reinterpret them as deltas. After the configured ten-step segment, new images and state are acquired and a new action chunk is requested. This loop continues until the terminal task condition or a stop condition is reached.

4.2.19 Control, computation, and safety boundary

The policy is not the safety controller. Before motion is enabled, a dry run verifies camera freshness and ordering, state/action dimensions, normalization statistics, checkpoint identity,

language instruction, finite model outputs, network communication, joint and gripper limits, workspace bounds, and emergency-stop access.

During motion, the run stops on a missing image, stale state, malformed packet, non-finite output, excessive target, workspace violation, communication timeout, robot protective stop, emergency stop, or manual stop. Reset is performed only after policy execution has stopped and the shared workspace is clear. The power socket remains unpowered. These checks maintain a deterministic boundary around the probabilistic learned policy.

4.3 Detailed system design

4.3.1 Workcell layout

The two FR3 arms are mounted upright on opposite sides of a shared tabletop region. Task fixtures are placed inside the verified overlap of the two practical workspaces. The D435i is fixed above the shared region. One D405 is attached to each wrist. Each arm carries one custom gripper. The two GELLO leaders remain outside the follower workcell and are used by the operator during demonstration collection.

The lower row of Figure 4–1 records the selected physical arrangement: dual FR3 arms, a top view, wrist views, and custom grippers. The final setup extends the depicted single GELLO symbol to two leader assemblies. Exact base spacing, camera height, camera pitch, and fixture coordinates were not included in the supplied implementation record; these values should be added to a future dimensioned workcell drawing rather than inferred from the conceptual image.

4.3.2 Dual-arm robotic platform

Both arms use the same 16-value bilateral ordering defined in Equations 4–4 and 4–5. The left and right controllers remain independent low-level systems, but they are driven from one policy output and one episode timeline. Each arm can perform a large scene-preparation motion or a precise manipulation motion, depending on the demonstrated task role.

The redundant seventh joint is important in the shared workcell because end-effector pose alone does not determine a unique arm posture. Demonstrations implicitly select postures that provide camera visibility, avoid fixtures, and reduce inter-arm interference. The low-

level controller remains responsible for executing the admissible targets; the learned model does not directly output torque.

4.3.3 GELLO leader assemblies

Each leader measures seven arm channels and one gripper-related channel for its assigned follower. The operator does not bear follower-arm load through the leader; it is a position-input and command device. Mechanical looseness, servo zero error, cable interference, or incorrect joint signs can degrade demonstration quality, so the two leader assemblies are inspected and checked independently before bilateral collection. The component-level construction and quantities are recorded in Appendix 2, rather than repeated in the system-level design description.

4.3.4 End-effector and mechanical components

Each gripper uses two screw-mounted compliant fingers. The mounting screws must be tightened consistently, and the fingers must be installed with matching orientation. Before a run, the operator checks for tearing at rib roots, permanent bending, loose mounting holes, surface contamination, and insufficient opening range.

The task fixtures prevent uncontrolled object drift while retaining the manipulation required by the task. The can and socket box must still be pushable into the intended central region. The motherboard must be graspable and movable by the left gripper but sufficiently supported to allow the right gripper to extract the memory module. The tray must remain fixed so that successful placement is not confused with tray motion.

4.3.5 Vision and sensing system

The D435i mount is fixed relative to the workcell and is positioned to keep all source and destination regions in view. The two D405 mounts move with the wrists and are oriented so that the gripper contact region appears in the close-range image during approach. Cables are routed to avoid entering the gripper opening or restricting wrist motion.

The logical camera mapping is fixed:

$$\begin{aligned} \text{overhead} &\equiv \text{D435i}, \\ \text{left_wrist} &\equiv \text{left D405}, & \text{right_wrist} &\equiv \text{right D405}. \end{aligned}$$

The recorder and policy use these identities consistently. The three cameras provide RGB to the model; their depth and inertial capabilities are not part of the current learned observation.

4.3.6 Demonstration data-collection system

The collection system starts the two robot interfaces, both gripper interfaces, three camera publishers, the ROS 2 bridge, and the LeRobot recorder. The bridge assembles the latest required fields into a packet. The recorder begins a named task episode only after all required fields are available.

The operator executes the complete instruction and then ends the episode. The episode is retained only if the three image streams decode correctly, state and action arrays have the required width, and the demonstration matches the intended task. Manual workcell reset is performed between episodes. The same task instruction is used during collection and later deployment so that language conditioning does not depend on an undocumented paraphrase.

4.3.7 Data processing pipeline

Processing has separate visual and numeric branches. The visual branch decodes the three RGB streams, converts them to channel-first tensors, preserves aspect ratio during resizing, pads to 224×224 , and normalizes to $[-1, 1]$. The numeric branch reads the 16-dimensional state and action, applies the training statistics, and pads the model-side representation to 32 dimensions.

Episode metadata determines the task and split. The eight validation episode indices are excluded from training for every task. During data loading, the task-balanced sampler chooses among the three task datasets with equal probability. The language instruction, three images, state, and supervised action target are then passed to the policy training step.

4.3.8 Policy-training system

The training system loads `pi05_base`, the converted PyTorch weights, and the LeRobot v3.0 datasets. It uses batch size 4 on the RTX A6000. Optimizer state, learning-rate schedule, model state, step index, and validation quantities are saved at the configured checkpoints.

Validation runs every 5,000 steps. Each task is evaluated separately before the unweighted macro-average is calculated. This preserves per-task traceability and prevents a low loss on

the largest dataset from concealing a poor result on another task.

4.3.9 Robot control and deployment system

The deployment system loads one checkpoint for all three instructions. The selected instruction and current observation are sent to $\pi_{0.5}$. A 50-step action chunk is returned, and the executor uses the first ten steps. The target ordering, absolute-action interpretation, 5 Hz cadence, and camera mapping are identical to training.

The left and right arm targets are separated only after policy output validation. The gripper commands are expressed as absolute opening widths in metres. After each executed segment, the queue is not blindly extended with the rest of the old 50-step chunk; the policy is queried again using updated images and state.

4.3.10 Hardware–software interfaces

Table 4–5 Detailed hardware–software interfaces in the final system

Information	Source	Destination	Implemented contract
Left leader joints	Left GELLO	Left follower interface	Seven leader positions mapped to left FR3 joint order, signs, offsets, and permitted range.
Left leader gripper	Left GELLO	Left gripper interface	One leader channel mapped to absolute gripper opening width.
Right leader joints	Right GELLO	Right follower interface	Seven leader positions mapped to right FR3 joint order, signs, offsets, and permitted range.
Right leader gripper	Right GELLO	Right gripper interface	One leader channel mapped to absolute gripper opening width.
Left measured joints	Left FR3 controller	ROS 2 bridge / recorder / executor	Seven current joint positions in state indices 0–6.
Left measured gripper	Left gripper	ROS 2 bridge / recorder / executor	Current opening width in metres at state index 7.
Right measured joints	Right FR3 controller	ROS 2 bridge / recorder / executor	Seven current joint positions in state indices 8–14.
Right measured gripper	Right gripper	ROS 2 bridge / recorder / executor	Current opening width in metres at state index 15.

Information	Source	Destination	Implemented contract
Overhead RGB	RealSense D435i	Bridge / recorder / policy	RGB 424×240 at 5 Hz; stored as $3 \times 240 \times 424$, then resized and padded to $3 \times 224 \times 224$.
Left-wrist RGB	Left RealSense D405	Bridge / recorder / policy	Same tensor contract under the fixed left-wrist feature name.
Right-wrist RGB	Right RealSense D405	Bridge / recorder / policy	Same tensor contract under the fixed right-wrist feature name.
Assembled packet	ROS 2 bridge	LeRobot recorder	Same-packet images and measured state; adjacent-packet arm action label.
Episode record	LeRobot recorder	Dataset v3.0	Three videos, state, absolute action, task instruction, episode and frame indexing.
Training sample	Dataset and sampler	$\pi_{0.5}$ trainer	Equal-probability task sampling; training episodes only.
Language instruction	Task interface	$\pi_{0.5}$ policy	One of the three documented task instructions matching the prepared scene.
Policy observation	Preprocessor	$\pi_{0.5}$ policy	Three normalized $3 \times 224 \times 224$ images and a padded numeric state whose first 16 entries are valid.
Predicted action	$\pi_{0.5}$ policy	Robot adapter	50-step chunk; first 10 steps consumed per query.
Left arm command	Robot adapter	Left FR3 controller	Absolute joint targets from action indices 0–6.
Left gripper command	Robot adapter	Left gripper	Absolute opening width from action index 7.
Right arm command	Robot adapter	Right FR3 controller	Absolute joint targets from action indices 8–14.
Right gripper command	Robot adapter	Right gripper	Absolute opening width from action index 15.
Stop conditions	Cameras, bridge, adapter, robot controllers, operator	Both robot command paths	Stop on stale/missing data, invalid action, timeout, workspace or joint violation, protective stop, emergency stop, or manual stop.

4.3.11 Task definitions and bilateral role allocation

Table 4–6 Language instructions, bilateral roles, and terminal conditions

Task	Language instruction	Demonstrated arm roles	Terminal completion condition
T1	First, push the can to the middle, then pick up the black ring with the handle, put it over the can, and set it down.	Left arm pushes the can; right arm grasps the handled ring and places it.	The can is repositioned and the released ring remains stably over the can.
T2	Push the box with the power socket to the middle, then pick up the plug and insert it into the socket.	Right arm pushes the socket box; left arm grasps and inserts the plug.	The socket box is repositioned and the plug remains inserted without manual assistance.
T3	Pull the computer motherboard closer, then remove the memory module and place it in the yellow tray.	Left arm grips and pulls the motherboard; right arm removes and places the memory module.	The memory module is fully removed and remains inside the yellow tray.

4.3.12 System operation

4.3.12.1 Pre-run checks

The operator confirms the checkpoint, instruction, object arrangement, camera mapping, gripper condition, network status, and robot status. Both arms are returned to the approved start region. The socket is verified to be unpowered. The shared workspace is cleared of unrelated objects, and the emergency stop remains accessible.

4.3.12.2 Collection run

The leader offsets and channel directions are checked. Recording begins only after all three cameras and all bilateral state/action fields are present. The operator executes the complete task through the two GELLO leaders. The episode is stopped, reviewed, accepted or rejected, and then the scene is reset.

4.3.12.3 Policy dry run

The policy is first queried without enabling robot motion. The returned shape, finite values, absolute-action interpretation, camera ordering, and initial target relative to the current

robot state are inspected. A dry run that fails any check is not promoted to physical execution.

4.3.12.4 Autonomous run

The executor enables the approved command path, consumes ten actions from each predicted chunk, and re-observes the workcell. The operator does not manually assist a counted trial. The trial ends when the documented terminal condition is met or a stop condition occurs. Outcome, stop reason, checkpoint, instruction, and video record are retained for Chapter 5 evaluation.

4.3.13 Parts and components

Table 4–7 retains only the system-level composition needed to understand the implementation. Manufacturer model details, leader subcomponents, and the complete 22-item parts record are provided in Appendix 2, Table 2–5.

Table 4–7 System-level component summary

System group	Quantity	Function
FR3 follower systems	2	Left- and right-arm manipulation with low-level robot control and protection.
Custom grippers	2	Task-compatible grasping with four compliant TPU fingers in total.
RGB camera views	3	One fixed overhead view and two wrist-local views.
GELLO leaders	2	Bilateral demonstration input, one leader per follower arm.
Training and bridge compute	1 system	Dataset recording, packet assembly, policy training, validation, and deployment.
Task-object and fixture sets	3	Ring/can, plug/socket, and motherboard/memory-module tasks.
Safety and storage resources	1 set	Deterministic stopping, supervision, dataset storage, checkpoints, and logs.

4.4 Manufacturing and implementation plan

4.4.1 Implementation scope

The project reuses laboratory robot arms, controllers, cameras, computing resources, and network infrastructure. Custom manufacturing is concentrated in the compliant fingers, GELLO structural parts, camera mounts, and task fixtures. Software implementation covers collection, packet assembly, LeRobot conversion, preprocessing, multi-task sampling, $\pi_{0.5}$ training, validation, inference, action adaptation, and run logging.

The full economic value of the workcell is not equal to the incremental cash expenditure. The FR3 arms, cameras, GPU, and safety equipment are in-kind laboratory resources. The DR3 budget recorded no additional out-of-pocket expense for the proof of concept. The expanded budget below preserves that statement while separating unpriced in-kind assets from incremental fabrication cost.

4.4.2 Materials and manufacturing processes

4.4.3 Gripper fabrication and inspection

The finger CAD is prepared for TPU 64D printing with the mounting face and rib structure preserved. The selected print orientation must avoid weak inter-layer loading at the mounting holes and rib roots. After printing, supports and stringing are removed without cutting the compliant ribs. The overall envelope, paired symmetry, hole condition, and free deformation are inspected before assembly.

The two fingers on one gripper are installed with matching orientation and adequate clearance through the commanded opening range. A manual opening-and-closing check verifies that neither finger contacts the camera mount or wrist cable. The assembly is rejected if a rib is torn, the base is permanently warped, a screw cannot retain the finger, or the two contact surfaces are visibly mismatched.

4.4.4 GELLO assembly and calibration

Each leader is assembled from the FPX330-H101 structure, printed links and brackets, seven M288 actuators, one M077 actuator, controller board, and 5 V power supply. Joint axes are checked for free movement before wiring. Cables are routed so that the full demonstrated

Table 4–8 Materials, manufacturing processes, and critical considerations

No.	Part or resource	Material or source	Process or tool	Critical consideration
1	FR3 arms and controllers	Laboratory equipment	Upright installation and interface configuration	Base rigidity, shared workspace, controller limits, network isolation, emergency stop
2	Compliant fingers	TPU 64D	Fused-filament 3D printing, support removal, inspection	Print orientation, rib integrity, screw-hole quality, paired geometry, permanent deformation
3	Gripper attachment	Existing gripper and printed/adapted interfaces	Mechanical assembly with screws	Alignment, thread engagement, consistent tightening, opening clearance
4	GELLO structures	FPX330-H101 and printed parts	Additive manufacturing and modular assembly	Joint-axis alignment, low friction, cable strain relief, repeatable zero offsets
5	Camera mounts	Laboratory or printed brackets	Rigid mounting and cable routing	Working distance, FOV, gripper visibility, occlusion, cable interference
6	Task fixtures	Printed or manually assembled fixture parts	Additive manufacturing and bench assembly	Repeatable initial pose, pushability where required, no mains power, stable destination
7	Software environment	LeRobot, PyTorch, ROS 2, project modifications	Version-controlled configuration and scripted execution	Commit traceability, feature names, absolute-action setting, dataset/model compatibility
8	Dataset and checkpoints	Recorded demonstrations and training outputs	Validation, indexing, backup, and integrity checks	Episode-level split, three-view completeness, task balance, reproducible checkpoint selection

range does not pull on connectors or bias the leader posture.

After electrical assembly, servo identifiers, zero positions, directions, and range mappings are verified one channel at a time. The left and right leaders are calibrated independently because a mirrored workcell does not imply identical signs and offsets. The final mapping is stored with the collection configuration so that later episodes can be traced to the leader calibration used.

4.4.5 Camera and fixture implementation

The overhead mount is installed so that the complete source, interaction, and destination regions remain visible. Each D405 is mounted within its intended 7–50 cm close-range operating envelope during manipulation. The wrist view is checked at approach, contact, and retreat postures, not only at the robot start pose.

Task fixtures are placed in the verified shared workspace. The can, socket box, and motherboard must permit the instructed preparation motion, while their target regions remain visible. The yellow tray is fixed against accidental displacement. The socket fixture is electrically isolated and unpowered. Reference markings or stops are used where necessary to make scene reset repeatable.

4.4.6 Software implementation

The software stack is configured in independent stages:

1. robot, gripper, camera, and GELLO device interfaces;
2. ROS 2 state and command topics;
3. bridge packet assembly and freshness checks;
4. LeRobot Dataset v3.0 episode recording;
5. image preprocessing and 16-to-32-dimensional model padding;
6. episode-level split and equal-probability task sampling;
7. $\pi_{0.5}$ fine-tuning and macro-loss validation;
8. unified-checkpoint policy inference; and
9. bilateral absolute-action adaptation and safety checks.

The configuration records LeRobot version 0.5.2, the project commit, local modifications, base-checkpoint source, training steps, learning-rate schedule, camera feature names,

`use_relative_actions=false`, action chunk size, and `n_action_steps`. These values are treated as part of the reproducible implementation rather than left as implicit code defaults.

4.4.7 System integration sequence

Integration proceeds from independent hardware checks to full learned control. First, each arm and gripper is checked without the policy. Second, all three cameras are checked under their final logical names. Third, each GELLO leader is mapped to its follower. Fourth, bilateral teleoperation is tested at low speed with an empty workspace. Fifth, complete episodes are recorded and decoded. Sixth, a training batch is inspected for camera order, numeric order, and action alignment. Seventh, the policy is run without opening the robot command path. Finally, supervised physical trials are enabled.

This sequence isolates failures. A camera-identity error should be found before training; a leader offset should be found before collecting a large dataset; a state/action index error should be found before robot execution; and a policy-output error should be found during the dry run.

4.4.8 Budget

Table 4–9 separates the reported incremental cash cost from the laboratory equipment, compute, and fabrication resources supplied in kind. This distinction avoids presenting an in-kind research workcell as a zero-value system.

Table 4–9 Expanded implementation budget and resource status

No.	Component, service, and quantity	Provider	Cash cost	Status and accounting note
1	Two FR3 arms, controllers, and safety equipment (2 sets)	Laboratory	CNY 0 incremental	Existing in-kind assets; acquisition value not supplied
2	D435i, two D405 cameras, and mounts (3 cameras)	Laboratory	CNY 0 incremental	Existing in-kind assets; mount material not separately recorded
3	RTX A6000 training workstation and storage (1 set)	Laboratory	CNY 0 incremental	Existing shared compute; energy and depreciation not allocated
4	GELLO leader assemblies (2)	Laboratory / project resources	CNY 0 incremental	DR3 records availability; itemized acquisition value not supplied
5	TPU 64D compliant fingers (4)	Student Innovation Center	CNY 0 incremental	Fabricated through available resources; material mass and print time not recorded
6	Task objects and fixtures (3 sets)	Laboratory / Student Innovation Center	CNY 0 incremental	Available; final fixture-material value not supplied
7	Software stack (1 configuration)	Open source and project code	CNY 0 licence cost	LeRobot, ROS 2, PyTorch, and project modifications
Reported additional out-of-pocket cost CNY 0				Excludes in-kind asset value, labour, energy, and depreciation

The CNY 0 value is therefore an incremental cash figure, not the total system value. A full economic budget would require the purchase or replacement value of two FR3 systems, two GELLO assemblies, three cameras, the GPU workstation, storage, safety hardware, print material, print time, labour, and energy. Those values were not included in the supplied DR3

or final implementation record and should be added only from invoices or the laboratory asset register.

4.4.9 Current implementation status and remaining documentation

The final implementation now includes dual-arm state and action handling, three RGB inputs, LeRobot Dataset v3.0 data, one jointly trained $\pi_{0.5}$ checkpoint, equal-probability three-task sampling, the defined episode-level validation split, macro flow-matching checkpoint selection, and a 5 Hz absolute-action deployment interface. The selected DR3 hardware modules and engineering drawings have been retained in this chapter.

Several quantities remain documentation gaps rather than model-design gaps. They include measured FR3 base poses, camera extrinsics and mounting distances, dimensioned fixture locations, ring and can clearances, and final socket insertion clearance and depth. They also include motherboard and memory-module clearances, measured gripper force–deflection behaviour, TPU print settings, print mass and time, full in-kind asset value, and measured end-to-end latency. These values should be added if the report is expected to make production-level tolerance, structural, timing, or economic claims.

Chapter 5 Validation results

5.1 Validation objectives and scope

The validation evaluates the final unified $\pi_{0.5}$ system described in Chapter 4. Its principal purpose is to determine whether one jointly trained checkpoint can execute the three documented bimanual tasks on the physical dual-FR3 workcell. The evaluation therefore separates two quantities that have different meanings:

1. the offline normalized flow-matching loss used to select a checkpoint; and
2. the end-to-end physical task success rate measured after deployment.

The first quantity measures agreement with held-out demonstration samples and is used only for model selection. The second measures whether a complete physical trial reaches the task-specific terminal condition in Table 4–6. A low validation loss is not treated as a substitute for physical task completion.

The physical evaluation covers the three bounded task configurations used during final implementation: ring placement over a can, plug insertion into a two-hole socket, and memory-module removal from a motherboard followed by placement in a yellow tray. Each task was evaluated in 25 counted trials, giving 75 physical trials in total. The results establish performance within these prepared laboratory scenes; they do not by themselves demonstrate open-world object generalization, arbitrary language understanding, or production-line reliability.

5.2 Checkpoint selection and deployment configuration

5.2.1 Selection criterion

Chapter 4 defines the checkpoint-selection procedure in Equation 4–13. Validation was performed every 5,000 optimizer steps. For each validation point, the normalized flow-matching losses of the ring, socket, and RAM datasets were first calculated separately and then combined as the unweighted macro-average

$$\mathcal{L}_{\text{macro}} = \frac{\mathcal{L}_{\text{ring}} + \mathcal{L}_{\text{socket}} + \mathcal{L}_{\text{ram}}}{3}.$$

The checkpoint with the lowest recorded $\mathcal{L}_{\text{macro}}$ was selected. Under this criterion, the best checkpoint occurred at 95,000 training steps. The model selected for physical deployment was therefore the 95,000-step unified checkpoint rather than the final 100,000-step training state.

This selection preserves equal task importance. The ring dataset contains more frames than the other two datasets, but neither the training sampler nor the checkpoint-selection metric gives it greater task-level weight. The numerical per-task and macro loss values at 95,000 steps were not included in the supplied final result record and are therefore not invented in this chapter.

5.2.2 Deployment hardware

Training was performed on the RTX A6000 system documented in Chapter 4. Physical deployment used an NVIDIA GeForce RTX 5070 Ti. The deployed policy used the same three image inputs, 16-dimensional bilateral state, absolute 16-dimensional action representation, language-instruction interface, and `use_relative_actions=false` setting as the training configuration. It predicted a 50-step action chunk and the executor consumed ten steps before requesting an updated prediction.

The deployment-GPU model is recorded to distinguish training compute from inference compute. End-to-end inference latency, GPU utilization, and the achieved policy-query period were not measured in the supplied evaluation record, so the nominal 5 Hz action cadence and approximately two-second executed chunk remain interface settings rather than measured latency results.

5.3 Physical trial protocol and success definition

For each trial, the prepared scene corresponded to one of the three fixed language instructions in Table 4–6. The same 95,000-step checkpoint was used for every task. Task identity was provided through the language instruction; separate task-specific checkpoints were not loaded.

A trial was counted as an end-to-end success only when its complete terminal condition was achieved:

- for the ring task, the can was repositioned and the released ring remained over the can;

- for the socket task, the socket box was repositioned and the plug remained inserted in the two-hole socket; and
- for the RAM task, the memory module was fully removed from the motherboard and remained inside the yellow tray.

Failures were categorized by the first task stage that prevented the terminal condition. Partial completion was retained for failure analysis but was not counted as end-to-end success.

The current result record specifies the number of trials and their final outcomes, but it does not yet specify the reset tolerance, object-position randomization range, trial timeout, lighting variation, or elapsed completion time. Consequently, the results below are reported as descriptive success rates for the bounded setup rather than as estimates of performance over a formally defined distribution of initial conditions.

5.4 Overall physical results

For task k , the end-to-end success rate is

$$R_k = \frac{N_k^{\text{success}}}{N_k^{\text{trial}}} \times 100\%. \quad (5-1)$$

Table 5–1 reports the resulting task-level values.

Table 5–1 End-to-end physical success rates using the 95,000-step unified checkpoint

Task	Successful	Failed	Total	Success rate
Ring over can	21	4	25	84%
Plug into socket	13	12	25	52%
Memory-module removal	25	0	25	100%
All physical trials	59	16	75	78.7%

Because every task contains the same number of physical trials, the micro-average over all 75 trials and the unweighted macro-average of the three task success rates are both

$$\frac{59}{75} \times 100\% = \frac{84\% + 52\% + 100\%}{3} \approx 78.7\%.$$

This aggregate value is higher than the 70% task-execution target in Table 2–2. However, the aggregate conceals a large task difference. The RAM and ring tasks individually exceed

70%, whereas the socket task reaches only 52%. The final system therefore does not satisfy the task-execution threshold uniformly across all three tasks.

5.5 Task-specific results and failure analysis

5.5.1 Memory-module removal and tray placement

The RAM task succeeded in all 25 counted trials, giving an observed success rate of 100%. In the successful sequence, the left arm pulled and stabilized the motherboard while the right arm removed the memory module and transferred it to the yellow tray. No end-to-end failure was recorded in the 25-trial set.

This result shows that the selected demonstration data, bilateral role allocation, camera observations, compliant grippers, and unified checkpoint were sufficient for the evaluated motherboard configuration. The 25-out-of-25 result should nevertheless be interpreted as performance within the tested setup. A finite trial set with no observed failure does not establish a zero failure probability, and the current evaluation does not quantify sensitivity to motherboard pose, memory-module type, connector retention force, lighting, or camera displacement.

5.5.2 Ring placement over the can

The ring task succeeded in 21 of 25 trials, corresponding to 84%. Four trials failed. Table 5–2 separates the observed failure modes.

Table 5–2 Observed failure modes in the ring task

Failure mode	Count	Share of ring failures
Ring not successfully placed over the can	1	25%
Handle grasp failed because the ring or handle was skewed relative to the grasp approach	2	50%
Handle grasp failed because the ring base obstructed the approach	1	25%
Total	4	100%

Three of the four failures occurred before secure transport because the right gripper did not successfully acquire the ring handle. The remaining failure occurred at the ring-over-

can placement stage. The observed distribution therefore identifies handle acquisition as the dominant bottleneck in this task, rather than the initial can-pushing operation.

The two skew-related failures indicate sensitivity to the relative orientation between the handle and the demonstrated grasp approach. The obstruction failure shows that the ring base can remove the approach clearance required by the custom fingers. These observations motivate additional demonstrations around rotated or displaced handle configurations, explicit reset tolerances, and inspection of wrist-camera visibility near the ring base. They also provide a concrete basis for evaluating whether a revised finger profile or a more vertical pre-grasp approach would improve robustness; these changes are proposed from the observed failures and were not evaluated in the present trial set.

5.5.3 Plug grasping and two-hole socket insertion

The socket task succeeded in 13 of 25 trials, corresponding to 52%. Twelve trials failed. Table 5–3 reports the two observed failure categories.

Table 5–3 Observed failure modes in the socket task

Failure mode	Count	Share of socket failures
Left gripper did not acquire or retain the plug	2	16.7%
Plug grasped, but two-hole insertion was inaccurate	10	83.3%
Total	12	100%

The dominant failure is constrained insertion rather than plug acquisition. Ten of the twelve failures reached the stage at which the plug had been grasped but did not enter the two-hole socket accurately enough to satisfy the terminal condition. This is consistent with the tolerance analysis in Section 4.2.6: the insertion requires both lateral alignment of two pins or prongs with their corresponding holes and sufficiently small angular error through the insertion motion.

The result does not by itself identify a single root cause. Possible contributors include residual socket-box pose variation after the right-arm push, plug pose variation inside the compliant left gripper, limited visual information at the contact point, and the temporal interval over which ten predicted actions are executed before replanning. These are engineering hypotheses inferred from the failure stage, not measured causal conclusions. Confirming

them would require trial-level video review, socket and plug geometry measurements, and preferably logging of the wrist view and commanded joint targets immediately before contact.

5.6 Cross-task interpretation

The results demonstrate that one 95,000-step $\pi_{0.5}$ checkpoint can produce successful physical executions for all three language-selected tasks without loading separate policies. Performance is nevertheless task dependent. The RAM task was fully successful in the observed trial set, the ring task was largely successful but sensitive to handle acquisition, and the socket task remained limited by precision insertion.

The failure distribution also separates gross bimanual sequencing from local contact precision. No failure category in the supplied record was assigned to choosing the wrong arm role or loading the wrong task-specific checkpoint. The observed failures instead occur at local grasp or placement/insertion stages. This supports the feasibility of the unified bilateral action interface while showing that unified task representation does not remove the need for task-specific contact robustness.

The current three scenes are visually distinct and each scene is paired with its expected instruction. Consequently, these trials validate deployment of a language-conditioned unified checkpoint, but they do not isolate how strongly the learned behavior depends on the language input. A separate instruction-sensitivity experiment using paraphrased, exchanged, or deliberately mismatched instructions would be required to distinguish language grounding from task recognition based primarily on scene appearance.

5.7 Validation limitations

Four limitations constrain interpretation of the reported percentages. First, the physical record does not yet define the distribution of initial object poses or the permitted reset tolerance. Second, trial duration and end-to-end inference latency were not recorded, so speed and real-time performance cannot be compared quantitatively with the design goals. Third, the evaluation contains one selected checkpoint and no ACT, single-task, or alternative-checkpoint baseline; it therefore demonstrates final-system performance but not the superiority of $\pi_{0.5}$ over the DR3 baseline. Fourth, the exact normalized validation losses at 95,000

steps are not yet included, so the checkpoint-selection decision is traceable by method and step but not yet by its numerical loss table or curve.

These limitations do not change the counted physical outcomes, but they define the boundary of the supported claims. The strongest supported conclusion is that the unified checkpoint executed all three scoped tasks on the physical dual-arm setup with task-dependent reliability. Broader claims about generalization, timing, language sensitivity, or production readiness require the additional measurements identified above.

5.8 Validation summary

The 95,000-step checkpoint, selected by the minimum task-balanced macro flow-matching loss, was deployed on an NVIDIA GeForce RTX 5070 Ti and evaluated in 75 physical trials. The memory-module task achieved 25/25 successes, the ring task achieved 21/25, and the socket task achieved 13/25. The combined result was 59/75, or 78.7%.

Failure analysis shows that ring failures were dominated by unsuccessful handle acquisition, while socket failures were dominated by inaccurate alignment and insertion into the two-hole socket. The RAM and ring tasks exceeded the 70% task-execution target, but the socket task did not. The validation therefore confirms successful unified multi-task deployment and identifies precision plug insertion as the principal remaining performance bottleneck.

Chapter 6 Discussion and conclusions

6.1 Discussion

This project developed a bimanual industrial manipulation system for representative 3C assembly operations. The final system integrates a dual-Franka Research 3 platform, GELLO-based demonstration collection, synchronized visual and proprioceptive data recording, task-compatible end effectors and fixtures, and an ACT-based policy deployment workflow. The design process remains traceable from customer requirements to engineering specifications, concept generation, concept selection, implementation, and validation.

A key outcome is that the bimanual manipulation task has been completed on the integrated real-robot setup. This confirms that the selected mechanical, sensing, data, and policy modules can operate together as a system rather than only as isolated components. The completed bimanual result also supports the original motivation of moving beyond single-arm demonstrations toward coordinated manipulation in a shared workcell.

The main limitation is that the present implementation remains a bounded laboratory proof of concept. The task setup, workspace, initial conditions, and fixtures are controlled. The ACT controller provides a deployable imitation-learning baseline, while language-conditioned checkpoint selection, unified multi-task VLA inference, autonomous planning, and generalized recovery remain outside the demonstrated scope.

6.2 Conclusions

The thesis presents an end-to-end engineering process for VLA-oriented bimanual industrial manipulation. Starting from customer requirements, the project defined measurable engineering specifications, generated modular solution concepts, selected a dual-FR3 and GELLO-based architecture, and implemented a data-driven manipulation pipeline using synchronized sensing and ACT policy deployment.

The completed system demonstrates that a dual-arm robot workcell can be integrated with teleoperation, dataset generation, learning-based control, and real-robot execution for the scoped bimanual manipulation task. The contribution is therefore both a working pro-

prototype and a traceable design methodology: the final design choices can be related back to the QFD requirements, while the validation structure provides a basis for reporting task success, perception quality, pose accuracy, failure handling, and recovery once the detailed data section is completed.

Overall, the project establishes a practical foundation for flexible 3C assembly research. It shows that bimanual manipulation can be achieved within a controlled laboratory workcell using the selected hardware and learning pipeline, while leaving task-specific details and quantitative key data to be documented separately.

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Documentation of Generated Concepts Appendix 1

Purpose and design evolution

This appendix preserves the alternatives generated during the DR3 concept-development process and records the implementation-stage change from an ACT baseline to the LeRobot $\pi_{0.5}$ policy. The alternatives were retained because they expose the principal engineering trade-offs in platform selection, teleoperation, sensing, end-effector design, and learned control. The update to $\pi_{0.5}$ does not invalidate the earlier concept generation; it documents how implementation evidence and the requirement for language-conditioned multi-task control changed the final policy decision.

Table 1–1 Evolution from the DR3 baseline to the final implementation

Design aspect	DR3 baseline	Final implementation
Robot platform	Dual FR3 selected; single-arm evidence used for staged risk reduction	Dual FR3 physical execution
Policy	ACT in LeRobot; task-specific checkpoint selection	$\pi_{0.5}$ in LeRobot with language-conditioned task input
Task scope	Ring placement and plug insertion used as staged cases	Ring over can, plug insertion, and memory-module removal
Language	Future extension	Runtime task instruction supplied to the VLA policy
Control scope	Vision–action imitation-learning baseline	Vision–language–action bimanual control

Module alternatives retained after concept generation

C1: Robot-platform alternatives

The selected platform is the dual-FR3 workcell because it provides a direct controller path within the available laboratory ecosystem and supports coordinated bimanual manipulation. A dual Flexiv platform remains attractive for force-aware contact tasks, but it would require a different controller, teleoperation interface, and data-integration path. An ALOHA-

style platform offers a lower-cost bimanual research route, but its mechanical capability and software ecosystem are less directly aligned with the existing FR3 workcell. The selected platform accepts higher integration effort in exchange for continuity with the available hardware and a direct route to the final dual-arm system.

C2: Teleoperation alternatives

GELLO was selected because its leader-arm structure provides an intuitive relationship between operator motion and the robot state/action trajectories recorded for learning^[21]. PICO 4 Ultra offers an XR-controller and hand-tracking route, but would add motion-mapping and calibration work^[29]. The Universal Manipulation Interface (UMI) provides a portable hand-held demonstration approach^[22]; it remains a useful research comparator but is less directly coupled to the fixed dual-FR3 joint-state pipeline.

C3: Data-collection alternatives

The selected concept uses synchronized multi-view RGB observations together with the state and action channels of both arms and grippers. The global view provides workspace and fixture context, while wrist or side views preserve information around grasping and contact regions. A reduced one-wrist-plus-global configuration lowers storage, calibration, and synchronization burden, whereas a proprioception-only configuration removes visual synchronization entirely. Both alternatives reduce the information available for language-grounded object selection, occlusion handling, and contact-region observation.

C4: End-effector alternatives

The standard Franka parallel gripper provides a direct robot interface and repeatable parallel-jaw motion, but offers limited passive adaptation to object geometry. A rigid 3D-printed gripper can be tailored to one component and can provide accurate locating features. The compliant finger concept retained from DR3 offers greater tolerance to small geometric and pose variation and can reduce concentrated contact forces during grasping and insertion. Because the final system contains three materially different tasks, the deployed end-effector configuration must be documented per task rather than assuming that one finger geometry is optimal for the ring handle, plug body, and memory module.

C5: Policy alternatives

ACT was the deployable DR3 baseline because it was integrated with the existing LeRobot demonstration workflow and could predict temporally structured actions from images and robot state^[23]. Diffusion Policy remains a credible visuomotor baseline for contact-rich manipulation^[30]. Neither model natively provides the same language-conditioned task interface required by the final three-task system.

The final implementation therefore uses $\pi_{0.5}$ through LeRobot^[12,15]. Compared with the ACT baseline, the selected VLA policy accepts a natural-language task description in addition to visual and proprioceptive observations. This supports a common interface for the three demonstrated tasks and aligns the deployed controller with the original VLA objective. The selection is an implementation decision; superiority over ACT or Diffusion Policy should only be claimed if controlled comparative trials are reported.

Generated task concepts

The final demonstrations instantiate three task concepts from the common bimanual VLA architecture:

1. workspace preparation followed by ring grasping and placement over a can;
2. workspace preparation followed by grasping, alignment, and plug insertion;
3. motherboard repositioning followed by memory-module removal and placement in a tray.

These tasks were retained because they collectively exercise object repositioning, language-grounded object selection, bimanual coordination, grasping, transport, constrained insertion, component extraction, and terminal placement. Their exact language instructions and completion conditions are listed in Table 4–6.

Hardware Specifications and Parts Records Appendix 2

Purpose

This appendix collects manufacturer reference parameters, component-level construction details, engineering drawings, and the complete parts record. These items support procurement, maintenance, replication, and hardware verification, but they are separated from Chapter 4 so that the main engineering analysis remains focused on design suitability, workspace, contact tolerance, sensing coverage, learning interfaces, and system operation.

Franka Research 3 reference parameters

Chapter 4 retains the four FR3 characteristics that directly justify the selected workcell: seven degrees of freedom, a 3 kg nominal payload, an 855 mm maximum reach, and link-side torque sensing in all seven axes. Table 2–1 preserves the broader manufacturer parameter set used during DR3^[24-26].

Figure 2–1 records the arm appearance, joint-axis convention, and principal link dimensions. Figure 2–2 preserves the complete DR3 parameter graphic. The workspace envelope remains in Figure 4–2 because it is used directly in the main shared-workspace analysis.

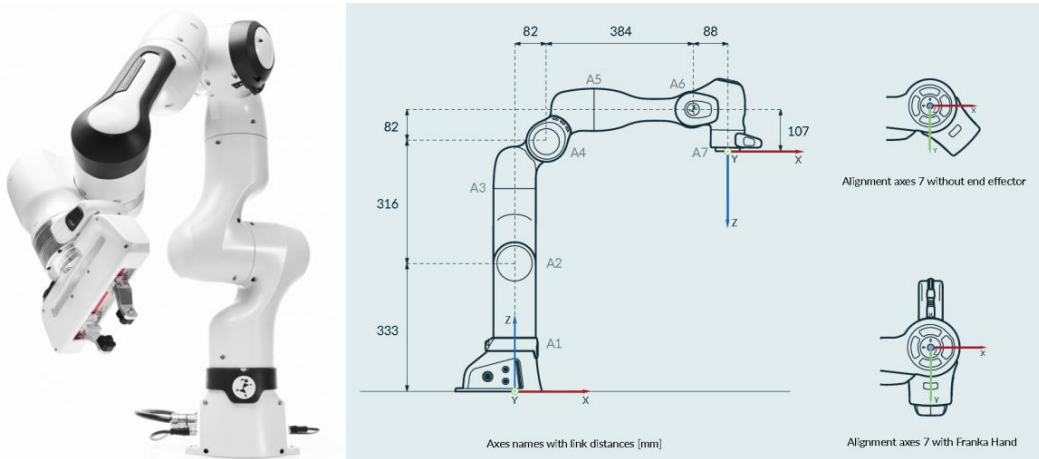


Illustration 2–1 FR3 arm, joint-axis convention, and principal kinematic dimensions retained as a hardware reference.

Table 2–1 FR3 manufacturer reference parameters used in the hardware record

Parameter	Reference value or interface
Degrees of freedom	7
Nominal payload	3 kg
Maximum reach	855 mm
Force/torque sensing	Link-side torque sensing in all seven axes
Joint position limits	A1 and A3: -166° to 166° ; A2: -105° to 105° ; A4: -176° to -7° ; A5: -165° to 165° ; A6: 25° to 265° ; A7: -175° to 175°
Mounting flange	DIN ISO 9409-1-A50
Installation position	Upright
Weight in the DR3 reference sheet	Approximately 17.8 kg; the value should be checked against the exact hardware and datasheet revision used in the laboratory
Protection rating	IP40
Ambient temperature	$+5^{\circ}\text{C}$ to $+45^{\circ}\text{C}$
Air humidity	20–80%, non-condensing
Primary interfaces	Ethernet (TCP/IP), control connector, end-effector connector, and safety-rated inputs
Project control boundary	High-level absolute targets are supplied through the project interface; low-level execution and manufacturer protection remain active

ARM			
Degrees of freedom	7	Interfaces	• ethernet (TCP/IP) for visual intuitive programming with Desk
Payload	3 kg		• safety-rated input for external enabling device
Maximum reach	855 mm		• 2 configurable safety-rated inputs for emergency stop devices, safeguards or other protective devices (OSSD devices via external OSSD converter connectable)
Force/Torque sensing	link-side torque sensor in all 7 axes		• hardware prepared for: 2x DI & 2x DO (24V, isolated, EN 61131-2 type 3 characteristics, 100 Hz sampling rate)
Joint position limits	A1, A3: -166/166 deg		• Control connector
	A2: -105/105 deg		• connector for end effector
	A4: -176/-7 deg	User Interfaces at the Arm's Pilot Grip	• integrated safety-rated guiding enabling switch
	A5: -165/165 deg		• guiding button
	A6: 25/265 deg		• guiding mode selector
	A7: -175/175 deg		
Mounting flange	DIN ISO 9409-1-A50	User Interfaces at the Arm's Pilot Disc	• status light
Installation position	upright		• Pilot mode selector
Weight	~ 17.8 kg		• arrow keys, teach, confirm, delete
Protection rating	IP40		
Ambient temperature ²	+5 °C to +45 °C		
Air humidity	20 – 80 % non-condensing		

Illustration 2–2 FR3 joint limits, mounting, environmental, and controller-interface parameters documented during DR3.

GELLO component-level construction

The final bilateral demonstration system uses one GELLO leader for each FR3 follower. Each leader supplies seven arm channels and one gripper-related channel. Table 2–2 lists the per-leader construction and the corresponding total for the dual-arm system.

Table 2–2 GELLO component-level construction

Component	Per leader	Dual total	Function
DYNAMIXEL XL330-M288-T	7	14	Measures the seven leader-arm joint channels
DYNAMIXEL XL330-M077-T	1	2	Provides the gripper-related input channel
ZNJ integrated control board	1	2	Reads and communicates the leader actuator states
Switched-mode power supply	1, 5 V 5 A	2	Supplies the leader electronics and actuator bus
FPX330-H101 structural kit	1 set	2 sets	Provides the modular leader structure
Printed links, brackets, and interfaces	1 set	2 sets	Completes the leader geometry, base, and Franka-compatible gripper input

The XL330 reference properties used during hardware selection are summarized in Table 2–3. They are component specifications rather than follower-robot requirements: the leader is used to measure and publish operator motion, not to reproduce the payload or torque capability of the FR3.

Table 2–3 DYNAMIXEL XL330 series reference properties

Property	Reference value
Input-voltage range	3.7–6 V
Recommended supply	5 V
Approximate dimensions	20 × 34 × 26 mm
Approximate mass	18 g per actuator
Supported control modes	Position, extended position, velocity, PWM, current, and current-based position modes
Selected variants	M288 for the seven arm joints and M077 for the gripper-related joint
Source	ROBOTIS XL330 documentation ^[31]

**Illustration 2–3 GELLO–Franka leader device retained as the component-level teleoperation reference.**

Compliant gripper drawing and fabrication reference

Each gripper assembly uses two identical TPU 64D fingers, giving four fingers in the dual-arm system. One finger has an overall bounding box of $123.36 \times 25.80 \times 40.00$ mm, corresponding to $L/T \approx 4.78$ and $L/H \approx 3.08$. It is screw mounted, with a reinforced attachment region and a tapered ribbed body.

Table 2–4 Compliant-finger fabrication reference

Item	Recorded design information
Material	TPU, Shore hardness 64D
Overall size	$123.36 \times 25.80 \times 40.00$ mm per finger
Quantity	Two fingers per gripper; four fingers for the dual-arm workcell
Attachment	Screw-mounted reinforced base
Contact geometry	Elongated tapered profile with repeated compliant ribs
Intended function	Passive accommodation of object-shape and pose variation during grasping and contact
Required inspection	Rib tearing, permanent bending, hole damage, paired-finger symmetry, screw retention, and opening clearance
Unreported quantities	Print orientation, infill, wall count, print mass and time, calibrated force–deflection curve, fatigue life, and contact load

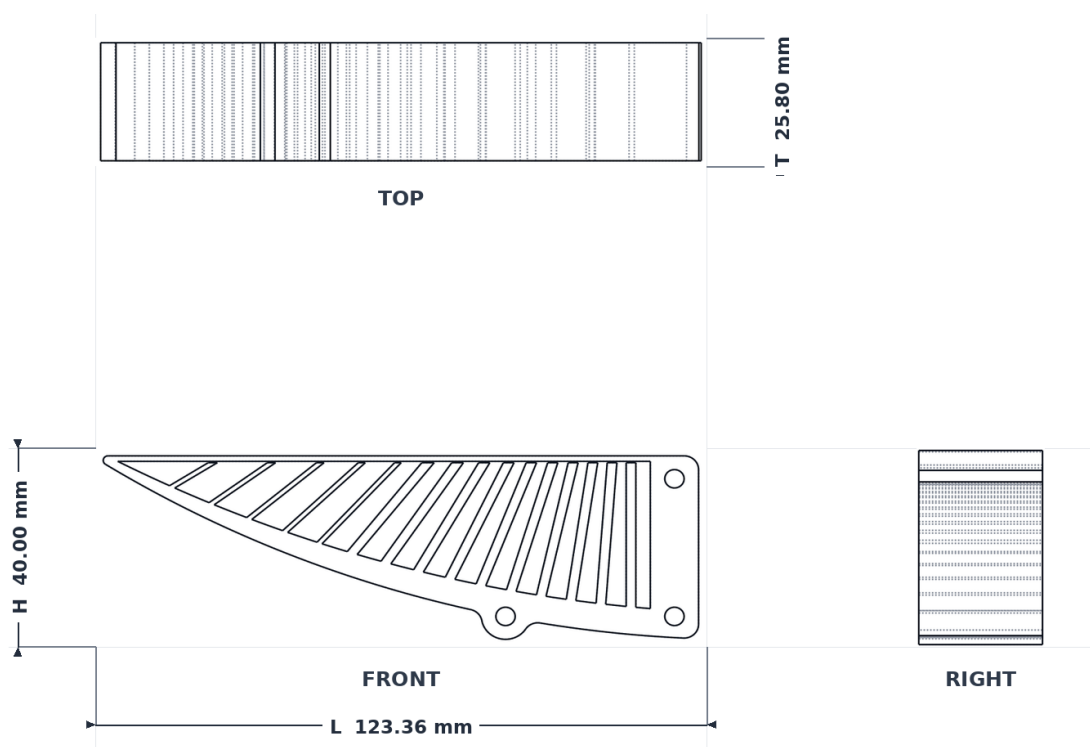


Illustration 2-4 Orthographic views of one TPU 64D compliant gripper finger and its overall dimensions.

Complete parts and components record

Table 2–5 preserves the complete system composition. Chapter 4 contains only a system-level summary so that component model numbers and procurement-oriented details do not interrupt the engineering argument.

Table 2–5 Complete parts and components record for the bimanual system

No.	Part or component	Specification	Qty.	Purpose
1	Franka Research 3 arm	7-DoF, 3 kg nominal payload, 855 mm maximum reach, joint torque sensing	2	Left and right follower manipulation
2	FR3 controller and robot interface	Manufacturer controller, network interface, low-level protection, FCI/ROS 2 integration	2 sets	State feedback and admissible joint-target execution
3	Custom gripper assembly	Project gripper with two screw-mounted compliant fingers	2	One end effector per arm
4	Compliant finger	TPU 64D, $123.36 \times 25.80 \times 40.00$ mm, tapered and ribbed	4	Object contact with passive geometric tolerance
5	Wrist camera	Intel RealSense D405, short-range RGB/depth device; RGB used by policy	2	Left- and right-wrist close-range observation
6	Overhead camera	Intel RealSense D435i; RGB used by policy	1	Global shared-workspace observation
7	GELLO leader assembly	7 arm channels plus 1 gripper channel	2	Bilateral human demonstration input
8	DYNAMIXEL XL330-M288-T	3.7–6 V actuator module; 5 V recommended	14	Seven arm joints per GELLO leader
9	DYNAMIXEL XL330-M077-T	3.7–6 V actuator module; 5 V recommended	2	Gripper-related channel on each GELLO leader
10	GELLO control board	ZNJ integrated control board	2	Reads and communicates leader joint positions
11	GELLO power supply	5 V, 5 A switched-mode supply	2	Leader-device electrical supply
12	GELLO structural kit	FPX330-H101 with project 3D-printed links and brackets	2 sets	Leader mechanism and mounting

No.	Part or component	Specification	Qty.	Purpose
13	Camera mounts and cable routing	Fixed overhead mount and two wrist mounts	1 set	Preserves camera identity, pose, and cable clearance
14	Training workstation	NVIDIA RTX A6000, 48 GB; physical GPU 1 used	1	Joint $\pi_{0.5}$ fine-tuning and validation
15	Network and ROS 2 bridge host	Project-configured communication and packet-assembly system	1 set	Connects robots, grippers, cameras, recorder, and policy
16	LeRobot software environment	LeRobot 0.5.2 with local changes; Dataset v3.0	1 config.	Recording, preprocessing, training, and deployment
17	Can and handled ring	Task 1 objects	1 set	Repositioning, grasp, and ring placement
18	Socket box and plug	Electrically unpowered insertion task fixture and matching plug	1 set	Repositioning and constrained insertion
19	Motherboard, memory module, and tray	Task 3 objects; yellow destination tray	1 set	Stabilization, extraction, transport, and placement
20	Task fixtures and reset references	Printed or manually assembled positioning aids	3 sets	Repeatable task initialization and terminal-state inspection
21	Safety equipment	Physical emergency stop, manufacturer protective functions, supervised operation	1 set	Deterministic stop and operator protection
22	Data storage	Storage for 119,650 frames, three-view video, checkpoints, and logs	1 set	Dataset and experiment traceability

Research Projects and Publications during Undergraduate Period

[1]

Acknowledgements

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A VISION-LANGUAGE-ACTION MODEL FOR DUAL-ARM ROBOTS IN 3C INDUSTRIAL PRODUCTION LINES

HCCI (Homogenous Charge Compression Ignition) combustion has advantages in terms of efficiency and reduced emission. HCCI combustion can not only ensure both the high economic and dynamic quality of the engine, but also efficiently reduce the NO_x and smoke emission. Moreover, one of the remarkable characteristics of HCCI combustion is that the ignition and combustion process are controlled by the chemical kinetics, so the HCCI ignition time can vary significantly with the changes of engine configuration parameters and operating conditions.